



Erik C. Andersen

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Johns Hopkins University**
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DATE UPDATED: June 9, 2026

MAJOR PROFESSIONAL INTERESTS

Evolutionary genetics; quantitative genetics; molecular genetics; parasitology; drug resistance

EDUCATION

- 2008-2013 Post-doctoral fellowship
Princeton University
Advisor: Dr. Leonid Kruglyak
- 2000-2008 Ph.D. in Biology
Massachusetts Institute of Technology (MIT), Cambridge, MA
Advisor: Dr. H. Robert Horvitz
Dissertation: The synthetic Multivulva genes and their suppressors regulate opposing cell fates through chromatin remodeling
- 1996-2000 B.S. in Biological Sciences with departmental honors
Stanford University, Stanford, CA
Advisor: Dr. Matthew P. Scott
Dissertation: *in vivo* analysis of *Drosophila* heart-tube formation

AWARDS, HONORS, AND FELLOWSHIPS

- 2022 - 2024 Fulbright Global Scholar (Chile, New Zealand, Taiwan), declined
- 2022 - 2024 Editorial Board Member of *Journal of Nematology*
- 2021 Distinguished Teaching Award, Weinberg College of Arts and Sciences
- 2020 - 2023 Editorial Board Member of *Gene*
- 2019 - 2022 Human Frontiers Science Program Award Recipient
- 2018 - 2023 National Science Foundation CAREER Award Recipient
- 2017 - 2020 Editor Board Member of *BMC Genomics*
- 2015 - 2019 American Cancer Society Research Scholar
- 2015 - 2017 March of Dimes Basil O'Connor Research Scholar
- 2015 - Editorial Board Member of *Trends in Genetics*
- 2014 - 2018 Pew Scholar in the Biomedical Sciences
- 2012 - 2013 Howard Hughes Medical Institute Post-doctoral Fellow
- 2011 - 2012 National Cancer Institute Post-doctoral Fellow, training grant T32-CA009528
- 2009 - 2011 Ruth L. Kirschstein National Research Service Award Recipient
- 2005 - 2006 Anna Fuller Cancer Graduate Research Fellowship
- 2000 Firestone Medal for Excellence in Undergrad. Research (top Biological Sciences researcher)
- 1999, 1998 Howard Hughes Medical Institute Summer Research Fellowship
- 1998 Stanford University Undergraduate research small grant recipient
- 1996-1998 Robert C. Byrd Honors Scholarship recipient

EMPLOYMENT

2025 - Vice Chair, Department of Biology, Johns Hopkins University
 2023 - Professor of Biology, Johns Hopkins University
 2022 - 2023 Professor of Molecular Biosciences, Northwestern University
 2022 - 2023 Professor of Cell and Developmental Biology, Feinberg School of Medicine, Northwestern University
 2020 - 2022 Associate Professor of Cell and Developmental Biology, Feinberg School of Medicine, Northwestern University
 2019 - 2022 Associate Professor of Molecular Biosciences, Northwestern University
 2013 - 2023 Preceptor for the Interdisciplinary Biological Sciences Graduate Program (IBiS)
 Resident member of the Chemistry of Life Processes Institute (CLP)
 Full Member of the Robert H. Lurie Comprehensive Cancer Center
 Member of Northwestern Institute on Complex Systems (NICO)
 2013 - 2019 Assistant Professor of Molecular Biosciences, Northwestern University
 2008 - 2013 Post-doctoral fellow, Princeton University, Advisor: Dr. Leonid Kruglyak
 2000 - 2008 Graduate student, Biology Department at Massachusetts Institute of Technology (MIT), Advisor: Dr. H. Robert Horvitz

RESEARCH SUPPORT**PRESENT**

2020 - 2031 National Institutes of Health (R01 AI153088) - NIAID
Discovery of novel benzimidazole resistance mechanisms
 Lead PI: Andersen, co-PIs: Fraser (UCSF), Schroeder (Cornell) Zamanian (UW Madison)

 2025 - 2028 Boehringer Ingelheim Research Support
Heterakis spumosa, a novel ascarid parasite model
 PI: Andersen

 2025 - 2028 Gordon and Betty Moore Foundation Award
 PI: Andersen

 2025 - 2028 National Science Foundation, EDGE CMT (2524223)
Defining the mechanistic roles of hyper-divergent regions in gene-by-environment interactions
 PI: Andersen

 2025 - 2029 National Institutes of Health (R01 AI182355)
Combining B. thuringiensis crystal proteins with small molecule anthelmintics to combat parasitic nematode resistance
 PI: Aroian (Univ. of Massachusetts Chan Med School), co-PI: Andersen

 2025 Genetic Resources Core Facility Discovery Initiative Award recipient
Optimization of ultra-long nanopore reads
 PI: Andersen

PAST

2024 - 2026 National Institutes of Health (R21 AI180805)
Investigating the genetics and genomics of antiparasitic drug responses in a poultry ascarid model
 PI: Andersen

- 2020 - 2026 National Institutes of Health (R01 AI153088) - NIAID
Discovery of novel benzimidazole resistance mechanisms
Lead PI: Andersen, co-PIs: Fraser (UCSF), Gilleard (U Calgary), Kaplan (U Georgia)
- 2022 - 2025 National Science Foundation, Capacity: Biological Collections (2224885)
Enhancement of the Caenorhabditis Natural Diversity Resource
co-PI: Andersen, co-PI: Tanny
- 2023 - 2025 US Poultry and Egg Association
Discovery of Novel Anthelmintic Resistance Genes to Reduce Production Loss Caused by Blackhead
PI: Andersen
- 2022 - 2024 National Institutes of Health (R21 OD030067) - Office of the Director
Genetic and genomic tools for C. briggsae research
Lead PI: Chamberlin (OSU), co-PI: Andersen
- 2018 - 2024 National Science Foundation CAREER Award (1751035)
Discovery of the molecular mechanisms underlying microevolution of phenotypic plasticity in a developmental trait
PI: Andersen
- 2018 - 2024 National Institutes of Health (R01 ES029930) - NIEHS
Discovery of conserved molecular mechanisms underlying population-wide variation in toxin responses
Lead PI: Andersen, co-PIs: Baugh (Duke), Rockman (NYU)
- 2019 - 2024 Human Frontiers Science Program Research Grant (RGP0001/2019)
The repeatability of the genetic mechanisms underlying behavioral evolution
Lead PI: Andersen, co-PIs: Brown (Imperial, MRC), Hodgins (Monash)
- 2022 - 2025 National Science Foundation, Research Experiences for Undergraduates (2150134)
No funding to REU leadership
REU Site: Quantitative Biology REU (QBREU) at Northwestern University
- 2018 - 2023 National Science Foundation (1764421) and Simons Foundation (597491)
Research Center for Mathematics of Complex Biological Systems
Understanding organismal growth and development through quantitative approaches
- 2022 - 2023 Les Turner ALS Center Pilot Grant
Creation of C. elegans ALS models incorporating natural variation
- 2019 - 2022 National Science Foundation, Collections in Support of Bio. Res. - Living Collections (1930382)
Enhancement of the Caenorhabditis Natural Diversity Resource
- 2017 - 2021 National Institutes of Health (R01 DK115690) - NIDDK
Large scale nutrigenetics and genomics in a tractable metazoan model
- 2015 - 2020 American Cancer Society Research Scholar Grant (127313-RSG-15-135-01-DDC)
Elucidating the genetic causes of variation in chemotherapy-based toxicity

- 2017 - 2020 National Institutes of Health (R21 AG053638) - NIA
High-throughput multi-modal analysis of natural variation in C. elegans healthspan
- 2016 - 2019 National Institutes of Health (R21 AI121836) - NIAID
Discovery and validation of avermectin resistance loci in free-living and parasitic nematodes
- 2014 - 2019 Pew Charitable Trusts, Scholars Program in the Biomedical Sciences
Elucidating the genetics of anthelmintic resistance in nematode-borne neglected tropical diseases
- 2015 - 2017 March of Dimes Basil O'Connor Starter Research Grant
Identification of hookworm anthelmintic resistance genes to ameliorate maternal and infant anemia
- 2014 - 2016 Chicago Biomedical Consortium, Catalyst Grant
Uncovering "missing heritability" in an experimentally tractable model organism

PUBLICATIONS

h-index=49, i10-index=91, link to Google Scholar page ([here](#))

Andersen lab graduate students denoted in *italics* and post-docs denoted in underline.

PREPRINTS:

117. Wang B, Moya ND, Tanny RE, Sauria MEG, *O'Connor LM*, Khorshidian A, McKeown R, Stevens L, Buchanan C, Crombie TA, *Dilks CM*, *Evans KS*, *Cook DE*, Zhang G, *Stinson LA*, Roberto NM, Lee D, *Zdraljevic S*, Gosse C, Gimond C, Chen ME, Dang VD, Wang J, Cutter AD, Rockman MV, Felix MA, Braendle C, and Andersen EC
Global genomic diversity of the selfing nematode *Caenorhabditis tropicalis* correlates with geography
bioRxiv, Posted Apr. 5, 2026
DOI: <https://doi.org/10.64898/2026.04.05.716573>
116. *Moya ND*, Wang B, Tanny RE, Sauria MEG, *O'Connor LM*, Khorshidian A, *McKeown R*, Gosse C, *Dilks CM*, Crombie TA, Zhang G, Rais E, Frezal L, Dan VD, Haryoso E, Devi MP, Gimond C, *Cook DE*, Hsu JC, *Shaver AQ*, *Zdraljevic S*, Richaud A, Wen T, Mehraj A, Sharanya H, Arulprakasam KR, Koury EJ, Roberto NM, *Schaye ES*, Singh V, Tarno H, Ailion M, Paaby AB, Zhao Z, Cutter AD, Wang J, Rockman MV, Felix MA, and Andersen EC
Caenorhabditis briggsae ancestral genomic hyper-diversity contrasts with globally distributed genome-wide haplotypes
bioRxiv, Posted Dec. 8, 2025
DOI: <https://doi.org/10.64898/2025.12.08.693002>
115. Clites BL, Frohock B, Koury EJ, Andersen EC, and Pierce JT (2024)
Natural variation in protein kinase D modifies alcohol sensitivity in *Caenorhabditis elegans*
bioRxiv, Posted Jun. 9, 2024
DOI: <https://doi.org/10.1101/2024.06.09.598102>
114. Fouad AD, Churgin MA, Hayden J, Xu J, Park JI, Liu A, Teng C, Sun H, Parrado M, Bowlin P, La Torre MD, Crombie TA, Sedore CA, Coleman-Hulbert AL, Johnson E, Phillips P, Andersen EC, and Fang-Yen C (2021)
High-throughput imaging of *Caenorhabditis elegans* aging using collective activity monitoring
bioRxiv, Posted Oct. 19, 2021
DOI: <https://doi.org/10.1101/2021.10.18.464905>

PEER-REVIEWED:

113. O'Connor LM, Moya ND, Jhaveri NS, Tanny RE, Khorshidian A, Lyu H, Chamberlin HM, Baird SE, and Andersen EC
A gap-free, telomere-to-telomere genome assembly for the *Caenorhabditis briggsae* reference strain AF16
Accepted at G3, June 12, 2026
bioRxiv, Posted Apr. 30, 2026
DOI: <https://doi.org/10.64898/2026.04.30.721887>
112. O'Connor LM, Singh A, Luallen RJ, Zhang G, and Andersen EC
Searching for vertically transmitted endosymbionts in over 4,000 wild strains of three self-fertilizing *Caenorhabditis* species
MicroPublication Biology. 2026 Jun 8: 10.17912/micropub.biology.002151
DOI: <https://doi.org/10.17912/micropub.biology.002151>
111. Shaver AO, McKeown R, Reyes Otero JM, and Andersen EC (2025)
Evaluating beta-tubulin variants as predictors of benzimidazole resistance across *Caenorhabditis* nematodes
PLoS Pathogens 22(6): e1014306.
DOI: <https://doi.org/10.1371/journal.ppat.1014306>
110. Saber S, Rifat MMI, Rahimi F, Snyder M, Singh A, Eickwort B, Newhall Y, Rajaei M, Saxena AS, Tanny RE, Katju V, Andersen EC, Zhou J, and Baer CF
Evolution of the rate, spectrum, and fitness effects of mutation under minimal selection in *Caenorhabditis elegans*
Genetics. 2026 Mar 26:iyag085.
DOI: <https://doi.org/10.1093/genetics/iyag085>
109. Collins JB and Andersen EC
Methods for single-pair *Ascaridia galli* genetic crosses
MicroPublication Biology. 2026 Mar 17: 10.17912/micropub.biology.002043
DOI: <https://doi.org/10.17912/micropub.biology.002043>
108. Crombie TA, McKeown R, Widmayer SJ, Shaver AO, Moya ND, Collins JB, Wit J, Tanny RE, Braendle C, Stevens L, van Sluijs L, Rockman MV, Sterken MG, Felix MA, and Andersen EC
Natural variation suggests candidate genes underlying *Caenorhabditis elegans* susceptibility to diverse toxicants
Toxicological Sciences. 2026 Mar. 2;209(3):kfag019.
DOI: <https://doi.org/10.1093/toxsci/kfag019>
107. Wei X, Richaud A, Tanny RE, Andersen EC, and Felix MA
Mutational divergence over years in local populations of the selfing nematode *Caenorhabditis elegans*
Genetics. 2026 Feb. 17: iyag043.
DOI: <https://doi.org/10.1093/genetics/iyag043>
106. Crombie TA, Pamminger T, Andersen EC, and Glaberman S (2026)
High-Throughput Toxicity Screening with *C. elegans*: Current Platforms, Key Advantages, and Future Directions
Environmental Science and Technology. 2026 Jan 20;60(2):1560-1576.
DOI: <https://doi.org/10.1021/acs.est.5c12562>

105. Peake AL, Jhaveri NS, Andersen EC, and Stinchcombe JR (2025)
High-throughput developmental assay of cold tolerance in *Caenorhabditis elegans*.
MicroPublication Biology. 2025 Nov 5: 10.17912/micropub.biology.001850.
DOI: <https://doi.org/10.17912/micropub.biology.001356>
104. Polk EA, Aqil A, Collins JB, Tanny RE, Andersen EC, Gokcumen O, and Ferkey DM (2025)
Wild Strains Reveal Natural Variation in *C. elegans* Avoidance Behaviors.
G3 (Bethesda). 2025 Oct 10;jkaf243.
DOI: <https://doi.org/10.1093/g3journal/jkaf243>
103. Collins JB, Choo R, Shaver AO, Schaye ES, Volpe T, Nunn L, Lighty ME, Niel KR, Frye EM, Zamanian M, and Andersen EC (2025)
A small-scale survey of fenbendazole resistance in *Ascaridia galli* and *Heterakis gallinarum*, two common ascarid parasites of poultry.
Poultry Science 2025 Sep 10;104(11):105808
DOI: <https://doi.org/10.1016/j.psj.2025.105808>
102. Jhaveri NS, Mastronardo MK, Collins JB, and Andersen EC (2025)
Development of a size-separation technique to isolate *Caenorhabditis elegans* embryos using mesh filters
PLoS ONE, 2025 Apr 24;20(4):e0318143
DOI: <https://doi.org/10.1371/journal.pone.0318143>
101. Jhaveri NS and Andersen EC (2025)
Fecundities of Hawaiian *Caenorhabditis briggsae* wild strains are not correlated with natural niche temperatures.
MicroPublication Biology, 2025 Jan 15; 2025:10.17912/micropub.biology.001356
DOI: <https://doi.org/10.17912/micropub.biology.001356>
100. Collins JB, Dilks CM, Hahnel SR, Rodriguez B, Fox BW, Redman E, Yu J, Cooke B, Sihuta K, Zamanian M, Roy PJ, Schroeder FC, Gillard JS, and Andersen EC (2024)
Naturally occurring variation in a cytochrome P450 modifies thiabendazole responses independent of beta-tubulin
PLoS Pathogens, 2025 Jan 14;21(1):e1012602
DOI: <https://doi.org/10.1371/journal.ppat.1012602>
99. Moya ND, Yan SM, McCoy RC, and Andersen EC (2024)
The long and short of hyper-divergent regions
Trends in Genetics, 2024 Nov 15
DOI: <https://doi.org/10.1016/j.tig.2024.11.005>
98. Shaver AO and Andersen EC (2024)
Integrating metabolomics into the diagnosis and investigation of anthelmintic resistance
Trends in Parasitology, 2024 Dec;40(12):1097-1106.
DOI: <https://doi.org/10.1016/j.pt.2024.10.020>
97. Zhang G, Felix MA, and Andersen EC (2024)
Transposon-mediated genic rearrangements underlie variation in small RNA pathways
Science Advances, 2024 Sep 20;10(38):eado9461.
DOI: <https://doi.org/10.1126/sciadv.ado9461>

96. Collins JB, Shaver AO, Schaye ES, Volpe T, Nunn LR, Zamanian M, and Andersen EC (2024)
Lack of correlation between parasite burden and key weight metrics in poultry infected with intestinal ascarids
MicroPublication Biology, 2024 Aug 7;2024:10.17912/micropub.biology.001197.
DOI: <https://doi.org/10.17912/micropub.biology.001197>
95. Crombie TA, Raja M, Saxena AS, Johnson LM, Saber S, Tanny RE, Ponciano JM, Andersen EC, Zhou J, and Baer CF (2024)
Direct inference of the distribution of fitness effects of spontaneous mutations from recombinant inbred *C. elegans* mutation accumulation lines
Genetics, 2024 Aug 14:iyae136.
DOI: <https://doi.org/10.1093/genetics/iyae136>
94. Collins JB, Stone S, Koury E, Paredes A, Shao F, Lovato C, Chen M, Shi R, Li A, Candal I, Al Moutaa K, Moya ND, and Andersen EC (2024)
Quantitative tests of albendazole resistance in beta-tubulin mutants
International Journal for Parasitology: Drugs and Drug Resistance, 2024 Jul 7;25(100556),
DOI: <https://doi.org/10.1016/j.ijpddr.2024.100556>
93. Yang H, Lee D, Kim H, *Cook DE*, Paik YK, Andersen EC, and Lee J (2024)
Glial expression of a steroidogenic enzyme underlies natural variation in hitchhiking behavior
PNAS 2024 Jul 3;121(28)e2320796121, DOI: <https://www.pnas.org/doi/10.1073/pnas.2320796121>
92. Knox J, Burns AR, Cooke B, Cammalleri SR, Kitner M, Ching J, Castelli JMP, Puumala E, Snider J, Koury E, Collins JB, Andersen EC, Stagljar I, Cowen LE, Laurens M, Zasada I, and Roy PJ (2024)
Cyprocide selectively kills nematodes via cytochrome P450 bioactivation
Nature Communications 2024 July 2;15(5529), DOI: <https://doi.org/10.1038/s41467-024-49738-4>
91. Shaver AO, Miller IR, Schaye ES, Moya ND, Collins JB, Wit J, Blanco AH, Shao FM, Andersen EJ, Khan SA, Paredes G, and Andersen EC (2024)
Quantifying the fitness effects of resistance alleles with and without anthelmintic selection pressure using *C. elegans*
PLoS Pathogens 2024 May 20;20(5):e1012245, DOI: <https://doi.org/10.1371/journal.ppat.1012245>
90. Lee H, Boor SA, Hilbert ZA, Meisel JD, Park J, Wang Y, *McKeown R*, Fischer SEJ, Andersen EC, and Kim DH (2023)
Genetic variants that modify the neuroendocrine regulation of foraging behavior in *C. elegans*
Science Advances 2024 Jun 14;10(20):eadk9481, DOI: <https://doi.org/10.1126/sciadv.adk9481>
89. Iyu H, Moya ND, Andersen EC, and Chamberlin HM (2024)
Gene duplication and evolutionary plasticity of *lin-12*/Notch gene function in *Caenorhabditis*
Genetics 2024 May 29:iyae064
DOI: <https://doi.org/10.1093/genetics/iyae064>
88. Wolstenholme AJ, Andersen EC, Choudhary S, Ebner F, Hartmann S, Holden-Dye L, Kashyap SS, Krücken J, Martin RJ, Midha A, Neysum P, Neveu C, Robertson AP, von Samson-Himmelstjerna G, Walker R, Wang J, Whitehead BJ, and Williams PDE
Getting around the roundworms: Identifying knowledge gaps and research priorities for the ascarids
Advances in Parasitology 2024 Feb 20
DOI: <https://doi.org/10.1016/bs.apar.2023.12.002>

87. Crombie TA, McKeown R, Moya ND, Evans KS, Widmayer S, LaGrassa V, Roman N, Tursunova O, Zhang G, Gibson S, Buchanan C, Roberto N, Vieira R, Tanny RE, and Andersen EC
CaeNDR, the *Caenorhabditis* Natural Diversity Resource
Nucleic Acids Research 2023 Oct 19; gkad887
DOI: <https://doi.org/10.1093/nar/gkad887>
86. Moya ND, Stevens L, Miller IR, Galindo JL, Bardas AD, Yeo C, Rozenich AJ, Xu M, Koh ESH, and Andersen EC (2023)
Novel and improved *Caenorhabditis briggsae* gene models generated by community curation
BMC Genomics 2023 Aug 25;24(1):486
DOI: <https://doi.org/10.1186/s12864-023-09582-0>
85. Wit J, Dilks CM, Zhang G, Kim Guisbert KS, Zdraljevic S, Guisbert E, and Andersen EC (2023)
Praziquantel inhibits *Caenorhabditis elegans* development and species-wide differences might be *cct-8*-dependent
PLoS ONE 2023 Aug 10;18(8):e0286473
DOI: <http://doi.org/10.1371/journal.pone.0286473>
84. Lee D, Fox BW, Palomino DCF, Panda O, Tenjo FJ, Evans KS, Stevens L, Koury E, Rodrigues PR, Kolodziej AR, Schroeder KF, and Andersen EC (2023)
Natural genetic variation in the pheromone production of *C. elegans*
The Proceedings of the National Academy of Sciences 2023 May 10;120(26)e2221150120
DOI: <https://doi.org/10.1073/pnas.2221150120>
83. Zhang G and Andersen EC (2023)
Interplay between polymorphic short tandem repeats and gene expression variation in *Caenorhabditis elegans*
Molecular Biology and Evolution 2023 Apr 4;40(4):msad067
DOI: <https://doi.org/10.1093/molbev/msad067>
82. Shaver AO, Wit J, Dilks CM, Crombie TA, Li H, Aroian RV, and Andersen EC (2023)
Variation in anthelmintic responses are driven by genetic differences among diverse *C. elegans* wild strains
PLoS Pathogens 2023 Apr 3;19(4):e1011285
DOI: <https://doi.org/10.1371/journal.ppat.1011285>
81. Venkatesan A, Jimenez Castro PD, Morosetti A, Horvath H, Chen R, Redman E, Dunn K, Collins JB, Fraser JS, Andersen EC, Kaplan RM, and Gilleard JS (2023)
Molecular evidence of widespread benzimidazole drug resistance in *Ancylostoma caninum* from domestic dogs throughout the USA and discovery of a novel β -tubulin benzimidazole resistance mutation
PLoS Pathogens 2023 Mar 2;19(3):e1011146
DOI: <https://doi.org/10.1371/journal.ppat.1011146>
80. Collins JB and Andersen EC (2022)
The turkey ascarid, *Ascaridia dissimilis*, as a model genetic system
International Journal for Parasitology 2022 Dec 19;S0020-7519(22)00177-1
DOI: <https://doi.org/10.1016/j.ijpara.2022.10.005>

79. Lesack K, Mariene GM, Andersen EC, and Wasmuth JD (2022)
Different structural variant prediction tools yield considerably different results in *Caenorhabditis elegans*
PLoS ONE 2022 Dec 30;17(12):e0278424
DOI: <https://doi.org/10.1371/journal.pone.0278424>
78. Nyaanga J, Shirman S, Mangan NM, and Andersen EC (2022)
Characterization of larval growth in *C. elegans* cuticle mutants
microPublication Biology 2022 Nov. 4, 2022:10.17912/micropub.biology.000662
DOI: <https://doi.org/10.17912/micropub.biology.000662>
77. Gibson S, Ness-Cohn E, and Andersen EC (2022)
Benzimidazoles cause lethality by inhibiting the function of *Caenorhabditis elegans* neuronal beta-tubulin
International Journal for Parasitology: Drugs and Drug Resistance 2022 Dec;20:89-96
DOI: <https://doi.org/10.1016/j.ijpddr.2022.10.004>
76. Shaver AO, Garcia BM, Gouveia GJ, Morse AM, Liu Z, Asef CK, Borges RM, Leach FE, Andersen EC, Amster IJ, Fernandez FM, Edison AS, and McIntyre LM (2022)
An anchored experimental design and meta-analysis approach to address batch effects in large-scale metabolomics
Frontiers Molecular Biosciences 2022 Nov 9;9:930204
DOI: <https://doi.org/10.3389/fmolb.2022.930204>
75. Zhang G, Wang Y, and Andersen EC (2022)
Natural variation in *C. elegans* short tandem repeats
Genome Research 2022 Oct;32(10):1852-1861
DOI: <https://doi.org/10.1101/gr.277067.122>
74. Widmayer SJ, Crombie TA, Nyaanga JN, Evans KS, and Andersen EC (2022)
C. elegans toxicant responses vary among genetically diverse individuals
Toxicology 2022 Aug 20; 479:153292
DOI: <https://doi.org/10.1016/j.tox.2022.153292>
73. Nyaanga J and Andersen EC (2022)
Linkage mapping reveals loci that underlie differences in *C. elegans* growth
G3 2022 Sep 30;12(10):jkac207
DOI: <https://doi.org/10.1093/g3journal/jkac207>
72. Wit J, Workentine ML, Redman E, Laing R, Stevens L, Cotton JA, Chaudry U, Ali Q, Andersen EC, Yeaman S, Wasmuth JD, and Gilleard JS (2022)
Genomic signatures of selection associated with benzimidazole drug treatments in *Haemonchus contortus* field populations
International Journal for Parasitology: Drugs and Drug Resistance 2022 Sep;52(10):677-689
DOI: <https://doi.org/10.1016/j.ijpara.2022.07.004>
71. Webster AK, Chitrakar R, Powell M, Chen J, Fisher K, Tanny RE, Stevens L, Evans KS, Antoshechkin I, Andersen EC, and Baugh LR
Using population selection and sequencing to characterize natural variation of starvation resistance in *Caenorhabditis elegans*
eLife 2022 Jun 21; 11:e80204
DOI: <https://doi.org/10.7554/eLife.80204>

70. Zhang G, Roberto NM, Lee D, Hahnel SR, and Andersen EC (2022)
The impact of species-wide gene expression variation on *Caenorhabditis elegans* complex traits
Nature Communications 2022 Jun 16; 13(1):3462
DOI: <https://doi.org/10.1038/s41467-022-31208-4>
69. Pallotto LM, Dilks CM, Park YJ, Smit RB, Lu B, Gopalakrishnan C, Gilleard JS, Andersen EC, and Mains PE (2022)
Interactions of *Caenorhabditis elegans* β -tubulins with the microtubule inhibitor and anthelmintic drug albendazole
Genetics 2022 Jul 30; 221(4):iyac093
DOI: <https://doi.org/10.1093/genetics/iyac093>
68. Fox BW, Ponomarova O, Lee YU, Zhang G, Giese GE, Walker M, Roberto NM, Na H, Reis-Rodriguez P, Curtis BJ, Kolodziej AR, Crombie TA, Zdraljevic S, Yilmaz LS, Andersen EC, Schroeder FC, and Walhout AJM (2022)
C. elegans as a model for inter-individual variation in metabolism
Nature 2022 Jul; 607(7919):571-577
DOI: <https://doi.org/10.1038/s41586-022-04951-3>
67. Crombie TA, Chikuturudzi C, Cook DE, and Andersen EC (2022)
An automated approach to quantify chemotaxis index in *C. elegans*
microPublication Biology 2022 May 26; 2022:10.17912/micropub.biology.000567
DOI: <https://doi.org/10.17912/micropub.biology.000567>
66. Widmayer SJ, Evans KS, Zdraljevic S, and Andersen EC (2021)
Evaluating the power and limitations of genome-wide association mapping in *C. elegans*
G3 2022 May 10; jkac114
DOI: <https://doi.org/10.1093/g3journal/jkac114>
65. Nyaanga J, Goss C, Zhang G, Ahmed HN, Andersen EJ, Miller IR, Rozenich JK, Swarthout IL, Vaughn JA, Mangan NM, Shirman S, and Andersen EC (2022)
Changes in body shape implicate cuticle stretch in *C. elegans* growth control
Cells and Development 2022 Apr 19;170:203780
DOI: <https://doi.org/10.1016/j.cdev.2022.203780>
64. Stevens L, Moya ND, Tanny RE, Gibson SB, Tracey A, Na H, Han Y, Chitrakar R, Dekker J, Walhout AJM, Baugh LR, and Andersen EC (2021)
Chromosome-level reference genomes for two strains of *Caenorhabditis briggsae*: an improved platform for comparative genomics
Genome Biology and Evolution 2022 Apr 10;14(4):evac042
DOI: <https://doi.org/10.1093/gbe/evac042>
63. Crombie TA, Tanny RE, Buchanan CM, Roberto NM, and Andersen EC (2022)
A highly scalable approach to perform ecological surveys of selfing *Caenorhabditis* nematodes
Journal of Visualized Experiments 2022 Mar 1;(181)
DOI: <https://doi.org/10.3791/63486>
62. Barlow I, Feriani L, Minga E, McDermott-Rouse A, O'Brien T, Liu Z, Hofbauer M, Stowers JR, Andersen EC, Ding SS, and Brown AEX (2022)
Megapixel camera arrays for high-resolution animal tracking in multiwell plates
Communications Biology 2022 Mar 23;5(1):253
DOI: <https://doi.org/10.1038/s42003-022-03206-1>

61. Crombie TA, Battlay P, Tanny RE, *Evans KS*, Buchanan CM, *Cook DE*, *Dilks CM*, *Stinson LA*, *Zdraljevic S*, Zhang G, Roberto NM, Lee D, Ailion M, Hodgins KA, and Andersen EC (2022)
Local adaptation and spatiotemporal patterns of genetic diversity revealed by repeated sampling of *Caenorhabditis elegans* across the Hawaiian Islands
Molecular Ecology 2022 Apr;31(8):2327-2347
DOI: <https://doi.org/10.1111/mec.16400>
60. Andersen EC and Rockman MV (2022)
Natural genetic variation as a tool for discovery in *Caenorhabditis* nematodes
Genetics 2022 Jan 4; 220(1):iyab156
DOI: <https://doi.org/10.1093/genetics/iyab156>
59. Gilbert KJ, *Zdraljevic S*, *Cook DE*, Cutter AD, Andersen EC, and Baer CF (2021)
The distribution of mutational effects on fitness in *Caenorhabditis elegans* inferred from standing genetic variation
Genetics 2022 Jan 4; 220(1):iyab166
DOI: <https://doi.org/10.1093/genetics/iyab166>
58. *Dilks CM*, Koury EJ, Buchanan CM, and Andersen EC (2021)
Newly identified parasitic nematode beta-tubulin alleles confer resistance to benzimidazoles
International Journal for Parasitology: Drugs and Drug Resistance 2021 Dec; 17:168-175
DOI: <https://doi.org/10.1016/j.ijpddr.2021.09.006>
57. Di Bernardo M, Crombie TA, *Cook DE*, and Andersen EC (2021)
easyFulcrum: An R package to process and analyze ecological sampling data generated using the Fulcrum mobile application
PLoS ONE 2021 Oct 6; 16(10):e0254293
DOI: <https://doi.org/10.1371/journal.pone.0254293>
56. *Nyaanga J*, Crombie TA, *Widmayer SJ*, and Andersen EC (2021)
easyXpress: An R package to analyze and visualize high-throughput *C. elegans* microscopy data generated using CellProfiler
PLoS ONE 2021 Aug 12; 16(8):e0252000
DOI: <https://doi.org/10.1371/journal.pone.0252000>
55. Rajaei M, Saxena AS, Johnson LM, Snyder MC, Crombie TA, Tanny RE, Andersen EC, Joyner-Matos J, and Baer CF (2021)
Mutability of mononucleotide repeats, not oxidative stress, explains the discrepancy between laboratory-accumulated mutations and the natural allele-frequency spectrum in *C. elegans*
Genome Research 2021 Sep;31(9):1602-1613
DOI: <https://doi.org/10.1101/gr.275372.121>
54. *Evans KS*, van Wijk MH, Andersen EC, and Sterken MG (2021)
From QTL to gene: *C. elegans* facilitates discoveries of the genetic mechanisms underlying natural variation
Trends in Genetics 2021 Jul 3; S0168-9525(21)00164-3
DOI: <https://doi.org/10.1016/j.tig.2021.06.005>

53. Gibson SB, Harper CS, Lackner LL, and Andersen EC (2021)
The *Caenorhabditis elegans* and *Haemonchus contortus* beta-tubulin genes cannot substitute for loss of the *Saccharomyces cerevisiae* beta-tubulin gene
microPublication 2021 Jun 20; 2021
DOI: <https://doi.org/10.17912/micropub.biology.000411>
52. Gouveia GJ, Shaver AO, Garcia BM, Morse AM, Rodriguez B, Park G, Andersen EC, Edison AS, McIntyre LM (2021)
Long-term metabolomics reference material
ACS Analytical Chemistry, 2021 Jul 6; 93(26):9193-9199
DOI: <https://doi.org/10.1021/acs.analchem.1c01294>
51. Zhang G, Mostad JD, and Andersen EC (2021)
Natural variation in fecundity is correlated with species-wide levels of divergence in *Caenorhabditis elegans*
G3 2021 Aug 7;11(8):jkab168
DOI: <https://doi.org/10.1093/g3journal/jkab168>
50. Wit J, Hahnel SR, Rodriguez BC, and Andersen EC (2021)
Natural variation in *Caenorhabditis elegans* responses to the anthelmintic emodepside
International Journal for Parasitology: Drugs and Drug Resistance 2021 Apr 17; 16:1-8
DOI: <https://doi.org/10.1016/j.ijpddr.2021.04.001>
49. Lee D, Zdraljevic S, Stevens L, Wang Y, Tanny RE, Crombie TA, Cook DE, Webster AK, Chirakar R, Baugh LR, Sterken M, Braendle C, Felix M-A, Rockman MV, and Andersen EC (2020)
Balancing selection maintains hyper-divergent haplotypes in *Caenorhabditis elegans*
Nature Ecology and Evolution 2021 Jun;5(6):794-807
DOI: <https://doi.org/10.1038/s41559-021-01435-x>
48. Hartman JH, Widmayer S, Bergemann C, King DE, Morton KS, Romersi RF, Jameson LE, Leung MCK, Andersen EC, Taubert S, and Meyer JN (2021)
Xenobiotic metabolism and transport in *Caenorhabditis elegans*
Journal of Toxicology and Environmental Health, Part B: Critical Reviews, 2021 Feb 17; 24(2):51-94
DOI: <https://doi.org/10.1080/10937404.2021.1884921>
47. Evans KS, Wit J, Stevens L, Hahnel SR, Rodriguez B, Park G, Zamanian M, Brady SC, Chao E, Introcaso K, Tanny RE, and Andersen EC (2021)
Two novel loci underlie natural differences in *Caenorhabditis elegans* abamectin responses
PLoS Pathogens, 2021 Mar 15; 17(3):e1009297
DOI: <https://doi.org/10.1371/journal.ppat.1009297>
46. Noble LM, Yuen J, Stevens L, Moya N, Persaud R, Moscatelli M, Jackson J, Braendle C, Andersen EC, Seidel HS, and Rockman MV (2021)
Selfing is the safest sex for *Caenorhabditis tropicalis*
eLife 2021 Jan 11;10:e62587
DOI: <https://doi.org/10.7554/eLife.62587>
45. Wit J, Dilks CM, and Andersen EC (2020)
Complementary Approaches to Understand Anthelmintic Resistance Using Free-Living and Parasitic Nematodes
Trends in Parasitology Dec 12: S1471-4922(20) 30323-30328
DOI: <https://doi.org/10.1016/j.pt.2020.11.008>

44. Evans KS and Andersen EC (2020)
The cadmium-responsive gene, *cdr-6*, does not influence *Caenorhabditis elegans* responses to exogenous zinc
MicroPublication Biology 2020 Sep 14;2020:10.17912/micropub.biology.000305
DOI: <https://doi.org/10.17912/micropub.biology.000305>
43. Hahnel SR, Dilks CM, Heising I, Andersen EC, and Kulke D (2020)
Caenorhabditis elegans in anthelmintic research - Old model, new perspectives
International Journal for Parasitology: Drugs and Drug Resistance 2020 Dec;14:237-248
DOI: <https://doi.org/10.1016/j.ijpddr.2020.09.005>
42. Evans KS, Zdravljec S, Stevens L, Collins K, Tanny RE, and Andersen EC (2020)
Natural variation in the sequestosome-related gene, *sqst-5*, underlies zinc homeostasis in *Caenorhabditis elegans*
PLoS Genetics 2020 Nov 11;16(11):e1008986
DOI: <https://doi.org/10.1371/journal.pgen.1008986>
41. Dilks CM, Hahnel SR, Sheng Q, Long L, McGrath PT, and Andersen EC (2020)
Quantitative benzimidazole resistance and fitness effects of parasitic nematode beta-tubulin alleles
International Journal for Parasitology: Drugs and Drug Resistance 2020 Dec;14:28-36
DOI: <https://doi.org/10.1016/j.ijpddr.2020.08.003>
40. Shaver AO, Gouveia GJ, Kirby PS, Andersen EC, and Edison AS (2020)
Culture and assay of large-scale mixed-stage *Caenorhabditis elegans* populations
Journal of Visualized Experiments 2021 May 5;(171)
DOI: <https://doi.org/10.3791/61453>
39. Na H, Zdravljec S, Tanny RE, Walhout AJM, and Andersen EC (2020)
Natural variation in a glucuronosyltransferase modulates propionate sensitivity in a *C. elegans* propionic acidemia model
PLoS Genetics 2020 Aug 28;16(8):e1008984
DOI: <https://doi.org/10.1371/journal.pgen.1008984>
38. Archer H, Deiparine S, and Andersen EC (2020)
The nematode *Caenorhabditis elegans* nematodes and the terrestrial isopod *Porcellio scaber* likely interact opportunistically
PLoS ONE 2020 Jun 26;15(6):e0235000
DOI: <https://doi.org/10.1371/journal.pone.0235000>
37. Evans KS and Andersen EC (2020)
The gene *scb-1* underlies variation in *Caenorhabditis elegans* chemotherapeutic responses
G3 2020 Jul 7;10(7):2353-2364
DOI: <https://doi.org/10.1534/g3.120.401310>
36. Zhao Y, Long L, Wan J, Biliya S, Brady SC, Lee D, Ojemakinde A, Andersen EC, Vannberg FO, Lu H, and McGrath PT (2020)
A spontaneous complex structural variant in *rcan-1* increases exploratory behavior and laboratory fitness of *Caenorhabditis elegans*
PLoS Genetics 2020 Feb 24;16(2):e1008606
DOI: <https://doi.org/10.1371/journal.pgen.1008606>

35. Bayat M, Tanny RE, Wang Y, Herden C, Daniel J, Andersen EC, Liebau E, Waschke DEJ (2020)
Effects of telomerase overexpression in the model organism *Caenorhabditis elegans*
Gene 2020 Mar 30;732:144367
DOI: <https://doi.org/10.1016/j.gene.2020.144367>
34. Crombie T, Zdravljec S, Cook DE, Tanny RE, Brady SC, Wang Y, Evans KS, Hahnel S, Lee D, Rodriguez BC, Zhang G, van der Zwaag J, Kiontke KC, and Andersen EC (2019)
Deep sampling of Hawaiian *Caenorhabditis elegans* reveals high genetic diversity and admixture with global populations
eLife 2019 Dec 3;8:e50465
DOI: <https://doi.org/10.7554/eLife.50465>
33. Brady SC and Andersen EC (2019)
An escape-room inspired game for genetics review
Journal of Biological Education 55:4, 406-417
DOI: <https://doi.org/10.1080/00219266.2019.1703784>
32. Daul AL, Andersen EC, and Rougvie AE (2019)
The *Caenorhabditis* Genetics Center (CGC) and the *Caenorhabditis elegans* Natural Diversity Resource
The Biological Resources of Model Organisms, CRC Press, Taylor and Francis Group
31. Webster A, Hung A, Moore B, Guzman R, Jordan J, Kaplan R, Hibshman J, Tanny RE, Cook DE, Andersen EC, and Baugh LR (2019)
Population selection and sequencing of *C. elegans* wild isolates identifies a region on chromosome III affecting starvation resistance
G3 2019 Oct 7;9(10):3477-3488
DOI: <https://doi.org/10.1534/g3.119.400617>
30. Lee D, Zdravljec S, Cook DE, Frezal L, Hsu JC, Sterken MG, Riksen JAG, Wang J, Kammenga JE, Braendle C, Felix MA, Schroeder FC, and Andersen EC (2019)
Selection and gene flow shape niche-associated variation in pheromone response
Nature Ecology and Evolution 2019 Oct;3(10):1455-1463
DOI: <https://doi.org/10.1038/s41559-019-0982-3>
29. Bernstein MR, Zdravljec S, Andersen EC, and Rockman MV (2019)
Tightly linked antagonistic-effect loci underlie polygenic demographic variation in *C. elegans*
Evolution Letters 2019 Sep 11;3(5):462-473
DOI: <https://doi.org/10.1002/evl3.139>
28. Gimond C, Vielle A, Silva Soares N, Zdravljec S, McGrath PT, Andersen EC, and Braendle C (2019)
Natural variation and genetic determinants of *Caenorhabditis elegans* sperm size
Genetics 2019 Oct;213(2):615-632
DOI: <https://doi.org/10.1534/genetics.119.302462>
27. Brady SC, Zdravljec S, Bisaga KW, Tanny RE, Cook DE, Lee D, Wang Y, Andersen EC (2019)
A novel gene underlies bleomycin-response variation in *Caenorhabditis elegans*
Genetics 2019 Aug;212(4):1453-1468
DOI: <https://doi.org/10.1534/genetics.119.302286>

26. Kim C, Kim J, Kim S, *Cook DE, Evans KS, Andersen EC, and Lee J* (2019)
Long-read sequencing reveals intra-species tolerance of substantial structural variations and new subtelomere formation in *C. elegans*
Genome Research 2019 Jun;29(6):1023-1035
DOI: <https://doi.org/10.1101/gr.246082.118>
25. *Zdraljevic S, Fox BW, Strand C, Panda O, Tenjo-Castano FJ, Brady SC, Crombie TA, Doench JG, Schroeder FC, and Andersen EC* (2019)
Natural variation in arsenic toxicity is explained by differences in branched chain amino acid catabolism
eLife 2019 Apr 8;8:e40260
DOI: <https://doi.org/10.7554/eLife.40260>
24. *Hahnel SR, Zdraljevic S, Rodriguez BC, Zhao Y, McGrath PT, and Andersen EC* (2018)
Extreme allelic heterogeneity at a *Caenorhabditis elegans* beta-tubulin locus explains natural resistance to benzimidazoles
PLoS Pathogens 2018 Oct 29;14(10):e1007226
DOI: <https://doi.org/10.1371/journal.ppat.1007226>
23. *Evans KS, Brady SC, Bloom JS, Tanny RE, Cook DE, Giuliani SE, Hippleheuser SW, Zamanian M, and Andersen EC.* (2018)
Shared genomic intervals underlie natural variation in diverse toxin responses
Genetics 2018 Dec;210(4):1509-1525
DOI: <https://doi.org/10.1534/genetics.118.301311>
22. *Zamanian M, Cook DE, Zdraljevic S, Brady SC, Lee D, Lee J, and Andersen EC* (2018)
Discovery of genomic intervals that underlie nematode responses to benzimidazoles
PLoS Neglected Tropical Diseases 2018 Mar 30;12(3):e0006368
DOI: <https://doi.org/10.1371/journal.pntd.0006368>
21. *Zdraljevic S and Andersen EC* (2017)
Natural diversity facilitates the discovery of conserved chemotherapeutic response mechanisms
Current Opinions in Genetics and Development 2017 Dec;47:41-47
DOI: <https://doi.org/10.1016/j.gde.2017.08.002>
20. *Zdraljevic S, Strand C, Seidel HS, Cook DE, Doench JG, and Andersen EC* (2017)
Natural variation in a single amino acid substitution underlies physiological responses to topoisomerase II poisons
PLoS Genetics 2017 Jul 12;13(7):e1006891
DOI: <https://doi.org/10.1371/journal.pgen.1006891>
19. *Lee D, Yang H, Kim J, Brady SC, Zdraljevic S, Zamanian M, Kim H, Paik Y, Kruglyak L, Andersen EC, and Lee J* (2017)
The genetic basis of natural variation in a phoretic behavior
Nature Communications 2017 Aug 17;8(1):273
DOI: <https://doi.org/10.1038/s41467-017-00386-x>
18. *Laricchia KM, Zdraljevic S, Cook DE, and Andersen EC* (2017)
The causes and consequences of natural variation in the distribution and abundance of transposable elements across the *Caenorhabditis elegans* species
Molecular Biology and Evolution 2017 Sep 1;34(9):2187-2202
DOI: <https://doi.org/10.1093/molbev/msx155>

17. Garcia-Gonzalez AP, Ritter AD, Shrestha S, Andersen EC, Yilmaz LS, Walhout AJM (2017)
Bacterial metabolism affects the *C. elegans* response to cancer chemotherapeutics
Cell 2017 Apr 20;169(3):431-441.e8
DOI: <https://doi.org/10.1016/j.cell.2017.03.046>
16. Cook DE and Andersen EC (2017)
VCF-kit: Assorted utilities for the variant call format
Bioinformatics 2017 May 15;33(10):1581-1582
DOI: <https://doi.org/10.1093/bioinformatics/btx011>
15. Mashock MJ, Zanon T, Kappell AD, Petrella LN, Andersen EC, Hristova KR (2016)
Copper oxide nanoparticles impact several toxicological endpoints and cause neurodegeneration in *Caenorhabditis elegans*
PLoS ONE 2016 Dec 2;11(12):e0167613
DOI: <https://doi.org/10.1371/journal.pone.0167613>
14. Evans KS, Zhao Y, Brady SC, Long L, McGrath PT, Andersen EC (2016)
Correlations of genotype with climate parameters suggest *Caenorhabditis elegans* niche adaptations
G3 2017 Jan 5;7(1):289-298
DOI: <https://doi.org/10.1534/g3.116.035162>
13. Cook DE, Zdraljevic S, Roberts JP, Andersen EC (2016)
CeNDR, the *Caenorhabditis elegans* Natural Diversity Resource.
Nucleic Acids Research 2017 Jan 4;45(D1):D650-D657
DOI: <https://doi.org/10.1093/nar/gkw893>
12. Cook DE, Zdraljevic S, Tanny RE, Seo B, Riccardi DD, Noble LM, Rockman MV, Alkema MJ, Braendle C, Kammenga JE, Wang J, Kruglyak L, Felix MA, Lee J, Andersen EC (2016)
The genetic basis of natural variation in *Caenorhabditis elegans* telomere length
Genetics 2016 Sep;204(1):371-83
DOI: <https://doi.org/10.1534/genetics.116.191148>
11. Large EE, Xu W, Zhao Y, Brady SC, Long L, Butcher RA, Andersen EC, McGrath PT (2016)
Selection on a Subunit of the NURF Chromatin Remodeler Modifies Life History Traits in a Domesticated Strain of *Caenorhabditis elegans*
PLoS Genetics 2016 Jul 28;12(7):e1006219
DOI: <https://doi.org/10.1371/journal.pgen.1006219>
10. Zamanian M and Andersen EC. (2016)
Prospects and challenges of CRISPR/Cas genome editing for the study and control of neglected vector-borne nematode diseases
FEBS 2016 Sep;283(17):3204-21
DOI: <https://doi.org/10.1111/febs.13781>
9. Farhadifar R, Ponciano JM, Andersen EC, Needleman DJ, Baer CF. (2016)
Mutation Is a Sufficient and Robust Predictor of Genetic Variation for Mitotic Spindle Traits in *Caenorhabditis elegans*
Genetics 2016 Aug;203(4):1859-70
DOI: <https://doi.org/10.1534/genetics.115.185736>

8. Sterken MG, Snoek LB, Kammenga JE, Andersen EC. (2015)
The laboratory domestication of *C. elegans*
Trends in Genetics 2015 May;31(5):224-31
DOI: <https://doi.org/10.1016/j.tig.2015.02.009>
7. Thompson OA, Snoek LB, Nijveen H, Sterken MG, Volkers RJM, Brenchley R, van't Hof A, Bevers RPJ, Cossins AR, Yanai I, Hajnal A, Schmid T, Perkins JD, Spencer D, Kruglyak L, Andersen EC, Moerman DG, Hillier LW, Kammenga JE, Waterston RH. (2015)
Remarkably divergent regions punctuate the genome assembly of the *Caenorhabditis elegans* Hawaiian strain CB4856
Genetics 2015 Jul;200(3):975-89
DOI: <https://doi.org/10.1534/genetics.115.175950>
6. Andersen EC, Shimko TC, Crissman JR, Ghosh R, Gerke JP, Seidel HS, Kruglyak L. (2015)
A powerful new quantitative genetics platform combining *Caenorhabditis elegans* high-throughput fitness assays with a large collection of recombinant strains
G3 2015 Mar 13;5(5):911-20
DOI: <https://doi.org/10.1534/g3.115.017178>
5. Farhadifar R, Baer CF, Valfort AC, Andersen EC, Muller-Reichert T, Delattre M, Needleman DJ. (2015)
Scaling, Selection, and Evolutionary Dynamics of the Mitotic Spindle
Current Biology 2015 Mar 16;25(6):732-740
DOI: <https://doi.org/10.1016/j.cub.2014.12.060>
4. Balla K, Andersen EC, Kruglyak L, Troemel E. (2015)
A wild *C. elegans* strain has enhanced epithelial immunity to a natural microsporidian parasite
PLoS Pathogens 2015 Feb 13;11(2):e1004583
DOI: <https://doi.org/10.1371/journal.ppat.1004583>
3. Etienne V, Andersen EC, Ponciano JM, Blanton D, Cadavid A, Joyner-Matos J, Matsuba C, Tabman B, Baer CF. (2015)
The Red Death Meets the Abdominal Bristle: Polygenic Mutation for Susceptibility to a Bacterial Pathogen in *Caenorhabditis elegans*
Evolution 2015 Feb;69(2):508-19
DOI: <https://doi.org/10.1111/evo.12585>
2. Shimko TC, Andersen EC. (2014)
COPASutils: an R package for reading, processing, and visualizing data from COPAS large-particle flow cytometers
PLoS ONE 2014 Oct 20;9(10):e111090
DOI: <https://doi.org/10.1371/journal.pone.0111090>
1. Andersen EC, Bloom JS, Gerke JP, Kruglyak L. (2014)
A variant in the neuropeptide receptor *npr-1* is a major determinant of *Caenorhabditis elegans* growth and physiology
PLoS Genetics 2014 Feb 27;10(2):e1004156
DOI: <https://doi.org/10.1371/journal.pgen.1004156>

Publications from research prior to my first tenure-track position:

- Felix MA, Jovelin R, Ferrari C, Han S, Cho YR, Andersen EC, Cutter AD, Braendle C. (2013)
Species richness, distribution and genetic diversity of *Caenorhabditis* nematodes in a remote tropical rainforest
BMC Evolutionary Biology, Jan 12;13:10.
- Ghosh R, Andersen EC, Shapiro JA, Gerke JP, Kruglyak L. (2012)
Natural variation in a chloride channel subunit confers avermectin resistance in *C. elegans*
Science, 335(6068): 574-578.
- Andersen EC*, Gerke JP*, Shapiro JA*, Crissman JR, Ghosh R, Bloom JS, Felix MA, Kruglyak L. (2012)
Chromosome-scale selective sweeps shape *Caenorhabditis elegans* genomic diversity
Nature Genetics, 44(3): 285-290. *equal contribution
- Andersen EC. (2011)
PCR-directed *in vivo* plasmid construction using homologous recombination in baker's yeast
Molecular Methods for Evolutionary Genetics, 772: 409-421. *Invited book chapter
- Raj A, Rifkin SA, Andersen EC, van Oudenaarden A. (2010)
Variability in gene expression underlies incomplete penetrance
Nature, 463(7283): 913-918.
- Bessler JB, Andersen EC, Villeneuve AB. (2010)
Differential localization and independent acquisition of the H3K9me2 and H3K9me3 chromatin modifications in the *Caenorhabditis elegans* adult germ line
PLoS Genetics, 6(1): e1000830
- Reddy KC*, Andersen EC*, Kruglyak L, and Kim DH. (2009)
A polymorphism in *npr-1* is a behavioral determinant of pathogen susceptibility in *C. elegans*
Science, 323(5912): 382-384. *equal contribution
- Andersen EC, Saffer AM, and Horvitz HR. (2008)
Multiple levels of redundant processes inhibit *Caenorhabditis elegans* vulval cell fates
Genetics, 179(4): 2001-2012.
- Andersen EC and Horvitz HR. (2007)
Two *C. elegans* histone methyltransferases repress *lin-3* EGF transcription to inhibit vulval development
Development, 134(16): 2991-2999.
- Reddien PW, Andersen EC, Huang M, and Horvitz HR. (2007)
DPL-1 DP, LIN-35 Rb, and EFL-1 E2F act with the MCD-1 Zinc-finger protein to promote programmed cell death in *C. elegans*
Genetics, 175(4): 1719-1733.
- Andersen EC, Lu X, and Horvitz HR. (2006)
C. elegans ISWI and NURF301 antagonize an Rb-like pathway in the determination of multiple cell fates
Development, 133(14): 2695-2704

Furlong EE, Andersen EC, Null B, White KP, and Scott MP. (2001)
 Patterns of gene expression during *Drosophila* mesoderm development
Science, 293(5535): 1629-1633

PROFESSIONAL SEMINARS

Departmental seminars and invited conference presentations (not including trainees):

- 2027 Dept. of Molecular and Cell Biology, Univ. of California - Berkeley, Berkeley, CA
 Parasitic Helminths: New Perspectives in Biology and Infection, Hydra, Greece (keynote, scheduled)
- 2026 Dept. of Microbiology, Immunology, and Tropical Medicine, George Washington Univ., Washington, DC
 Bronx High School of Science, New York NY (outreach lecture)
 16th International Conference of Parasitology, Montreal, Canada (two keynotes, scheduled)
 Society of Nematology, Baltimore, MD (keynote, scheduled)
 AAVP, Pre-meeting: Optimizing Parasiticides in the Face of Drug resistance (organizer, speaker)
 European Society of Nematologists, Egmond aan Zee, Netherlands (workshop keynote)
 Dept. of Biology, Univ. of Maryland, College Park, MD
- 2025 Dept. of Biology, Univ. of Buffalo, Buffalo, NY
 EMBL Workshop on Wild Frontiers of Model Organisms, Heidelberg, Germany
 Les Treilles Foundation meeting on nematode evolution, Nice, France
- 2024 Molecular genetics, genomics, microbiology, University of Strasbourg, Strasbourg, France
 Dept. of Biology, Lehigh University, Lehigh, PA
 Dept. of Biochemistry, Cellular, and Molecular Biology, Univ. of Tennessee, Knoxville, TN
 Dept. of Human Genetics, University of California - Los Angeles
 Evolution of *Caenorhabditis* and other nematodes, Vienna, Austria
 Poultry Science Dept., Univ. of Veterinary Medicine - Vienna, Vienna, Austria
 NIH, Laboratory of Parasitic Diseases, Bethesda, MD
 Carnegie Institute of Embryology, Baltimore, MD
 Baltimore Worm Club
- 2023 Chicago Area Worm Meeting (keynote)
 Zoom about Parasites (ZaP worms), virtual seminar
 Latin American Worm Meeting, Valparaiso, Chile
- 2022 Institute for Integrative Genome Biology, Univ. of California - Riverside, Riverside, CA
 International Congress of Toxicology, Maastricht, Netherlands
 Human Frontiers Research Symposium, Paris, France
 The *C. elegans* Metabolism, Aging, Pathogenesis, Stress and Small RNAs
 Genetics and Molecular Biology seminar series, University of North Carolina - Chapel Hill
 Dept. of Biology, Oklahoma University, Norman, OK
 Wellcome Evolutionary Systems Biology
- 2021 The Ascarid Research and Training Initiative, Lyon, France (Keynote)
 Bridging the Divide: International Worm Meeting (Keynote)
 66th Annual Meeting of the American Association of Veterinary Parasitologists (Keynote)
 Max Planck Institute for Evolutionary Biology, Plön, Germany
 Institut de Biologie Valrose, Nice, France
- 2020 American Society of Tropical Medicine and Hygiene, symposium on genetic crosses
 Dept. of Molecular Medicine, Univ. of Massachusetts Medical School, Worcester, MA
 Oakton Community College, Des Plaines, IL
 Chengdu Research Base of Giant Panda Breeding, Chengdu, People's Republic of China
- 2019 Netherlands Institute for Ecology (NIOO-KNAW), Wageningen, Netherlands
 Dept. of Genetics, Wageningen Univ. and Research, Wageningen, Netherlands
 Dept. of Nematology, Wageningen Univ. and Research, Wageningen, Netherlands
 Dept. of Genome Sciences, Univ. of Washington, Seattle, WA

- 2019 Institute of Molecular Biology, Academia Sinica, Taiwan
 Dept. of Genetics, Univ. of Georgia, Athens, GA
 Dept. of Infectious Diseases, Univ. of Georgia, Athens, GA
 Dept. of Biology, Georgia Institute of Technology, Atlanta, GA
 Ingram Cancer Center, Vanderbilt University, Nashville, TN
- 2018 Max Planck Institute for Developmental Biology, Tübingen, Germany
 Dept. of Evolutionary Ecology and Genetics, Christian-Albrechts-Universität, Kiel, Germany
 Berlin Seminar for Resistance Research, Freie Universität Berlin, Berlin, Germany
 Robert H Lurie Comprehensive Cancer Center, Northwestern University, Chicago, IL
 Dept. of Genetics, Washington University School of Medicine, St. Louis, MO
 Dept. of Genetics, University of Pennsylvania, Philadelphia, PA
 Dept. of Pathobiology, University of Pennsylvania, Philadelphia, PA
 Dept. of Microbiology, Seoul National University, Seoul, Korea
 Dept. of Biology, University of Oregon, Eugene, OR
 Dept. of Biological Sciences, University of Southern California, Los Angeles, CA
 Dept. of Cell Biology and Anatomy, Rosalind Franklin University, Chicago, IL
 New York Univ. Center for Genomics and Sys. Bio. Parasite Workshop, Abu Dhabi, UAE
 Department of Biology, Univ. of California - San Diego, San Diego, CA
 Department of Biology, Duke University, Durham, NC
 Donnelly Centre for Cellular And Biomolecular Research, University of Toronto, Toronto, ON
 Department of Molecular Biology and Genetics, Cornell University, Ithaca, NY
 Medical Research Council, London Institute of Medical Sciences, London, UK
 Department of Biology, Carnegie Mellon University, Pittsburgh, PA
 Host-Parasite Interactions, University of Calgary, Banff, Canada
 Florida Area Worm Meeting (Keynote), Florida Institute of Technology, Melbourne, FL
 Department of Biology, Skirball Institute, New York University Medical School, New York, NY
 Department of Biology, University of Minnesota, Minneapolis, MN
- 2017 Department of Biology, Indiana University, Bloomington, IN
 New York University Center for Genomics and Systems Biology Symposium in Abu Dhabi, UAE
- 2016 Midwest Quantitative Biology at Purdue University, West Lafayette, IN
 Molecular and Cellular Biology of Helminth Parasites X, Hydra, Greece
 Computational Research Day, Northwestern University, Evanston, IL
 Evolutionary Biology of *Caenorhabditis* and other nematodes (Keynote), CSHL, Cold Spring Harbor, NY
 Department of Genetics, University of Utah, Salt Lake City, UT
 Department of Biology, University of Iowa, Iowa City, IA
 Department of Biomedical Sciences, Iowa State University, Ames, IA
 Anthelmintics: Discovery to Resistance II, San Diego, CA
 Program in Systems Biology, University of Massachusetts Medical School, Worcester, MA
- 2015 Evolution seminar series, University of Wisconsin, Madison, WI
 Biotechnology Training Program, Northwestern University, Evanston, IL
 Department of Biology, Johns Hopkins University, Baltimore, MD
 Department of Biology, University of Maryland, College Park, MD
 Department of Pharmacology, Feinberg School of Medicine, Northwestern University, Chicago, IL
 Midwest Neglected Infectious Disease Meeting, Notre Dame University, South Bend, IN
 Quantitative genetics workshop, 20th International *C. elegans* meeting, UCLA, Los Angeles, CA
 Michigan Area Worm Meeting, van Andel Institute, Grand Rapids, MI
- 2014 Northwestern Institute on Complex systems, Northwestern University, Evanston, IL
 Fondation de Treilles: Revisiting the roles of phenotypic plasticity in evolution, Provence, France
 Biology Department, Marquette University, Milwaukee, WI
 Pharmacogenomics group, University of Chicago, Chicago, IL

Seminars before starting at first faculty position:

- 2013 Quantitative genetics workshop, 19th International *C. elegans* meeting, UCLA, Los Angeles, CA
Molecular Bioscience Department, Northwestern University, Evanston, IL
Program in Systems Biology, University of Massachusetts Medical School, Worcester, MA
- 2012 Biology Department, Dartmouth University, Hanover, NH
Human Genetics Department and Life Sciences Institute, University of Michigan, Ann Arbor, MI
Genetics Department, University of Georgia, Athens, GA
Biology Department, Case Western Reserve University, Cleveland, OH
Biology Department and BioDesign Institute, Arizona State University, Phoenix, AZ
Center for Computational and Integrated Biology, Rutgers University, Camden, NJ
Biology Department, University of Florida, Gainesville, FL
- 2011 Evolution workshop, 18th International *C. elegans* meeting, UCLA, Los Angeles, CA
Laboratory of Toxicology, NIEHS, Research Triangle Park, NC
- 2010 Institute for Evolutionary Biology Department, University of Edinburgh, Edinburgh, UK
- 2008 Featured talk at *C. elegans* Aging, Stress, and Pathogenesis meeting, Madison, WI
- 2000 Undergraduate research symposium, Stanford University, Stanford, CA

PEER REVIEW AND RELATED ACTIVITIES**Editorial Board:**

- 2022 - 2024 *Journal of Nematology*
2020 - 2023 *Genes*
2015 - *Trends in Genetics*

Associate Editor:

- 2017 - 2019 *BMC Genomics (Multicellular invertebrate genomics)*

Guest Associate Editor:

- PLoS Genetics, eLife, PLoS Pathogens, PLoS Neglected Tropical Disease, Genetics, G3*

Reviewing activity: Grants and fellowships

- 2026 *Ad hoc* reviewer ZRG1 BCTA-B(55)
Ad hoc reviewer ZRG1 BCTA-B(57)
University of California's Early Career Award Reviewer
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
NSF grant reviewer
Ad hoc reviewer NIH RM1 review panel (twice)
- 2025 NSF grant reviewer
Ad hoc reviewer BBSRC UK
Ad hoc reviewer European Science Foundation
Ad hoc reviewer Swiss National Science Foundation
Ad hoc reviewer NIH Neurotoxicology and Alcohol (NAL) Study Section
Ad hoc reviewer NIH Pathogenic Eukaryotes (PTHE) Study Section
- 2024 NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Ad hoc reviewer Fulbright Fellowship
International Research Olympiad Advisory Board
Ad hoc reviewer Japan Society for the Promotion of Science
Ad hoc reviewer BBSRC UK
Ad hoc reviewer Swiss National Science Foundation
- 2023 *Ad hoc* reviewer Dept. of Defense - Neurotoxicity
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q

- 2023 *Ad hoc* reviewer NIH ViCTER Award
Panel reviewer NSF EDGE
Ad hoc reviewer NIH NIDA Avenir Award
Ad hoc reviewer Dept. of Defense - Neurotoxicity
- 2022 NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Ad hoc BBSRC grant reviewer
Katholieke Universiteit Leuven grant reviewer
Ad hoc reviewer National Science Foundation (IOS)
Ad hoc reviewer NIH NIDA Avenir Award
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
- 2021 ERC Consolidator Grant reviewer (COI, declined)
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Canada Foundation for Innovation reviewer
NSF GRFP reviewer
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
- 2020 NIH Special Emphasis panel ZRG IFCN-C (02)
NIH Special Emphasis panel ZRG ETTN-N (02)
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
- 2019 *Ad hoc* reviewer for Wellcome Trust Early Career Grant
Panel reviewer on NASA, Flight and Ground Space Biology
Ad hoc reviewer for Agence Nationale de la Recherche
Ad hoc reviewer for Univ. of Wisconsin - Milwaukee Catalyst grant
Panel reviewer on NIH NIAID R13
Ad hoc reviewer National Science Foundation (CAREER)
Ad hoc reviewer for Swiss 3R Competence Centre
Ad hoc reviewer for Austrian Science Foundation
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
- 2018 *Ad hoc* reviewer for Alzheimer's Society
Ad hoc reviewer National Science Foundation (IOS)
Ad hoc reviewer National Science Foundation (CAREER)
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
Ad hoc reviewer for Bill and Melinda Gates Foundation
- 2016 ERC COST grant reviewer
- 2015 *Ad hoc* reviewer for National Toxicity Program, project assessment
- 2014 *Ad hoc* reviewer for Human Frontiers Science Program
Ad hoc reviewer for National Science Foundation (IOS)

PROFESSIONAL AFFILIATIONS AND SERVICE

Membership in Professional Societies:

Genetics Society of America
Society of Molecular Biology and Evolution (lifetime)
Society for Evolution (lifetime)
Society for Integrative and Comparative Biology
World Association for the Advancement of Veterinary Parasitology
American Association of Veterinary Parasitologists
Society of Nematologists

Mentorship and Diversity training:

- 2024 Johns Hopkins University, Center for the Improvement of Mentored Experiences in Research (CIMER), virtual
- Building Trust Through Effective Communication
 - Creating a Culture of Inclusion in your Lab
 - Aligning Expectations to Promote Independence
- 2022 Univ. of Wisconsin - Madison, Center for the Improvement of Mentored Experiences in Research (CIMER), in-person
- Arizona State Univ., Culturally Aware Mentoring, virtual
- 2021 Univ. of Southern California, Strategies for Equity-based Holistic Review in Graduate Admissions
- Northwestern Univ., Diversity, equity, and inclusion in hiring decisions
- 2020 NIH OITE training on Health and Wellness of Trainees

Professional service:

- 2027 International Worm Meeting, elected organizer, Kobe, Japan
- Anthelmintics VII Meeting, organizing committee, Queens University, Belfast
- 2026 AAVP, Pre-meeting workshop co-organizer
- Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
- 2025 International Worm Meeting, organizing committee/abstract reviewer
- 2024 *C. elegans* and other nematodes meeting organizer, IMBA, Vienna, Austria
- External thesis examiner member for Victor Loegler (Schacherer lab, Univ. of Strasbourg)
- C. elegans* Community Faculty Mentor, Genetics Society of America
- External thesis examiner member for Maria Mercado (Fraser lab, Univ. of Toronto)
- External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)
- 2023 External thesis examiner for Nikita Jhaveri (Gupta lab, McMaster Univ.)
- Worm Board *ex officio* member Nematode Genomes
- C. elegans* Community Faculty Mentor, Genetics Society of America
- External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)
- 2022 *C. elegans* Community Faculty Mentor, Genetics Society of America
- External thesis committee member for Leonor Gianechini (Moorhead lab, Univ. of Georgia)
- External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)
- External thesis committee member for Amanda Shaver (Edison lab, Univ. of Georgia)
- Worm Board *ex officio* member Nematode Genomes
- Organizing committee Anthelmintics V: Resistance and vaccines
- Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
- 2021 Worm Board *ex officio* member Nematode Genomes
- Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
- Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
- 2020 Worm Board *ex officio* member Nematode Genomes
- Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
- Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
- 2019 External thesis committee examiner for Yiru Wang (Kammenga lab, Wageningen Univ. and Research)
- External thesis committee member for Amanda Shaver (Edison lab, Univ. of Georgia)
- External thesis committee examiner for Aurian Garcia-Gonzalez (Walhout lab, UMMS)
- Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
- Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)

- 2018 Organizing committee for *C. elegans* dev., cell bio., and gene exp. meeting (Barcelona, Spain)
 Chair of the Natural Variation session, *C. elegans* dev., cell bio., and gene exp. meeting
 Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
 Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
 Organizing committee for Parasitic Nematodes meeting, NYU Abu Dhabi
 External thesis committee member for Victoria Vu (Fraser lab, University of Toronto)
 Poster judge, Northwestern Undergraduate Research Symposium
- 2017 Organizing committee for the 21st International *C. elegans* meeting
 Chair of the Evolution and Ecology parallel session, 21st International *C. elegans* meeting
- 2015 Organizing committee for the 20th International *C. elegans* meeting
 Poster judge, 20th International *C. elegans* meeting - Evolution and Genomics section
 Genetics Soc. of America Mentor Lunch, *Postdoc search*, 20th International *C. elegans* meeting
- 2014 Panelist, NUIN Post-doc Association, *Interviews and Start-up packages*
 Poster judge, Northwestern Undergraduate Research Symposium
 Panelist, Pathways to the Professoriate, *How to prepare for a job interview?*
- 2013 Poster judge, Northwestern Undergraduate Research Symposium
 Panelist, Bioscientist Freshman seminar; *How to find a research lab?*
 Poster judge, 19th International *C. elegans* meeting - Evolution and Genomics section

***C. elegans* community service and open-data:**

- 2016 - 2021 Creator and director of the *C. elegans* Natural Diversity Resource (CeNDR)
 This resource organizes and disseminates wild *C. elegans* strains, whole-genome sequence data, and enables genome-wide association mappings through a cloud-based service.
- 2021 - Creator and director of the *Caenorhabditis* Natural Diversity Resource (CaeNDR, [link](#))
 This resource expands on the on CeNDR after incorporating wild strains from *C. briggsae* and *C. tropicalis*. We also added new web tools and analysis frameworks to enable comparative and molecular quantitative genetics. All data are freely available to all.

TEACHING AND ADVISING

Undergraduate teaching:

- 2026 AS.020.383: *Advanced Genetics* (upcoming fall semester)
 AS.020.303: *Genetics* (summer semester, 11 students)
 AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone, Leah Froemming)
- 2025 **New course: AS.020.383: *Advanced Genetics* (fall semester, 23 students)**
 AS.020.314: *Biology of Disease* (fall semester, 180 students)
 AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone)
- 2024 AS.020.303: *Genetics* (spring semester, 465 students)
 AS.020.314: *Biology of Disease* (fall semester, 180 students)
 AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone)
- 2023 Biological Sciences 203: *Genetics and Evolution* (winter, 285 students)
 Biological Sciences 398: *Independent Research* (Fiona Shao, Alyssa Blanco)
 Biological Sciences 399: *Independent Research* (Andrea Phung, Cassia Yeo, Sharik Khan)
- 2022 **New course: Biological Sciences 203: *Genetics and Evolution* (winter, 379 students)**
 Biological Sciences 398: *Independent Research* (Andrea Phung)
- 2021 **Biological Sciences 393: *Genetic Analysis* (spring, 31 students)**
 Biological Sciences 399: *Independent Research* (Karan Gowda, Kailyn Parham, Katie Introcaso)

- 2020 **Biological Sciences 215: *Genetics and Molecular Biology* (spring, 492 students)**
Biological Sciences 399: *Independent Research* (Anna Derrick, Emily Jahn, Jake Mostad, Kailyn Parham)
- 2019 **Biological Sciences 393: *Biomedical Genetics* (spring, 19 students)**
Biological Sciences 398: *Tutorial in Biology* (Anna Derrick, Emily Jahn, Kailyn Parham)
Biological Sciences 399: *Independent Research* (Karol Bisaga, Grace Park, Jake Mostad)
- 2018 **Biological Sciences 393: *Genetic Analysis* (winter, 28 students)**
Biological Sciences 398: *Tutorial in Biology* (Karol Bisaga)
Biological Sciences 399: *Independent Research* (Karol Bisaga, Kimberly Collins, Selina Deiparine, Grace Park)
- 2017 **Biological Sciences 393: *Genetic Analysis* (spring, 22 students)**
Biological Sciences 398: *Tutorial in Biology* (Kimberly Collins)
Biological Sciences 399: *Independent Research* (Selina Deiparine, Samuel Hamilton, Grace Park)
- 2016 **Biological Sciences 393: *Genetic Analysis* (spring, 17 students)**
Biological Sciences 398: *Tutorial in Biology*
(Sarah Bier, Mattlyn Cordova, Selina Deiparine, Samuel Hamilton, Grace Park)
- 2015 *Guest Lecture: University of Wisconsin-Madison Biology 675 - Evolution seminar*
(fall, 8 students)
New course: Biological Sciences 393: *Genetic Analysis* (spring, 10 students)
Biological Sciences 398: *Tutorial in Biology* (Lautaro Cilenti)
Biological Sciences 399: *Independent Research* (Kreena Patel, Hillary Tsang)
- 2014 Biological Sciences 398: *Tutorial in Biology* (Mazeed Aro-Lambo, Kreena Patel, Hillary Tsang)

Graduate teaching:

- 2026 AS.020.683: *Advanced Genetics* (fall semester, upcoming)
- 2025 **New course: AS.020.683: *Advanced Genetics* (fall semester, 22 students)**
CMDB RCR course - *Collaborative Science* (one lecture, 26 students)
- 2024 CMDB RCR course - *Collaborative Science* (one lecture, 22 students)
- 2022 Driskill Graduate Program 430: *Genetic Analysis - Leveraging Big Data and Model Organisms*
(spring, one guest lecture, 12 students)
- 2021 **New course: Interdisciplinary Biological Sciences: *Introduction to R for biologists***
(fall, Five students)
Driskill Graduate Program 430: *Genetic Analysis - Leveraging Big Data and Model Organisms*
(spring, one guest lecture, 16 students)
- 2020 Interdisciplinary Biological Sciences 423: *Ethics of collaboration*
(fall, one guest lecture, 45 students)
- 2019 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility*
(summer, one guest lecture, 17 students)
- 2018 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility*
(summer, one guest lecture, 25 students)
Interdisciplinary Biological Sciences/Chemistry 416: *Practical Training in Chemical Biology Methods and Experimental Design*
(spring, five lectures, 10 students)
- 2017 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility*
(summer, one guest lecture, 15 students)
Interdisciplinary Biological Sciences/Chemistry 416: *Practical Training in Chemical Biology Methods and Experimental Design*
(spring, five lectures, 8 students)
Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility*
(spring, one guest lecture, 5 students)

- 2016 Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp*
(fall, 3 days, eight hours per day, 19 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp
- 2015 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology*
(fall, one guest lecture, 22 students)
Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp*
(fall, 3 days, eight hours per day, 22 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp
Interdisciplinary Biological Sciences 423: *Ethics of peer review*
(spring, one guest lecture, 41 students)
- 2014 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology*
(fall, one guest lecture, 16 students)
New course: Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp*
(fall, 3 days, eight hours per day, 16 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp
Interdisciplinary Biological Sciences 423: *Ethics of peer review*
(spring, one guest lecture, 42 students)
- 2013 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology*
(fall, one guest lecture, 24 students)

K-12 advising:

Esha Sharma, Richard Montgomery High School (2025-2026)
 Ariel Shi, Baltimore Polytechnic Institute (2025-2026)
 Cecilia Soko, Baltimore Polytechnic Institute (2025-2026)
 Arya Takalkar, Thomas Jefferson High School (2025)
 Colin Lily, Baltimore Polytechnic Institute (2024-2025)
 Jinghan (Jonathan) Xiao, Lake Forest Academy (2022-2023)
 Anwyn Zhou, New Trier High School (2022)
 Richelle Lee, Adlai E. Stevensen High School (2022)
 Preeti Rao, Adlai E. Stevensen High School (2022)
 Ned Koh, Lake Forest Academy (2021-2022)
 Yahya Junejo, Hinsdale Central High School (2020)
 Shanthi Hegde, Lambert High School (2020 - 2021)
 Aarnav Patel, Barrington High School (2020)
 Justine Rozenich, Saint Ignatius College Preparatory (2020-2022)
 Hannah Ahmed, Normal Community High School (2019-2020)
 Ally Bardas, New Trier High School (2019-2021)
Regeneron Science Talent Search Finalist 2021
 Sarosh Nagar, Glenbrook North High School (2018-2019 academic year)
 Britney Sun, Glenbrook North High School (summer 2018)
 Ethan Schonfeld, Glenbrook North High School (summer 2018)
 Lillian Tushman, Oak Park and River Forest High School (2016-2017 academic year)
Recipient of 3rd place All-Illinois Science Research Competition
 Caitlin Westerfield, Evanston Township High School (2015-2016 academic year)
 Matteo di Bernardo, Evanston Township High School (2015-2016 academic year)
Recipient Columbia University Scientific Scholars Fellowship, Recipient Fulbright Fellowship
 Ainsley Tran, Oak Park and River Forest High School (2015-2016 academic year)
 Lauren Mann, Oak Park and River Forest High School (2014-2015 academic year)
 Jacob Cruger, Latin School of Chicago (summers 2013, 2014)
 Gina Liu, Illinois Math and Science Academy (2013-2014 academic year)

Undergraduate advising:

Marie-Jose Jean Rohayem (2025 - , Class of 2028), Cell and Molecular Biology Major
Nora Ewis (2026, Class of 2027), Emory University
Leah Froemming (2026 - , Class of 2027), Cell and Molecular Biology Major
Olivia Jonathan (2026, Class of 2027), Boston University
Jack Weinstein (2025 - 2026, Class of 2027), Haverford College
Grayson Benson (2025 - , Class of 2027), Oberlin College
Kate Stone (2024 - , Class of 2027), Cell and Molecular Biology Major
2025 Johns Hopkins Summer Provost's Undergraduate Research Award (PURA) recipient
Rohan Shrishrimal (2024 - , Class of 2028), Cell and Molecular Biology Major
Rachel Choo (2023 - 2026, Class of 2026), Cell and Molecular Biology Major
Joyce Reyes-Otero (2024, Class of 2025), Industrial Microbiology Major, Univ. of Puerto Rico - Mayagüez
2024 Johns Hopkins Biology Summer REU recipient
Alyssa Blanco (2023 - 2024, Class of 2025), Biological Sciences Major, Northwestern University
2023 PBS Summer grant recipient
2024 Johns Hopkins Biology Summer REU recipient
Fiona Shao (2022 - 2023, Class of 2024), Biological Sciences Major
2023 Summer URG recipient
Gracie Paredas (2022 - 2023, Class of 2025), Biological Sciences Major
2023 Summer URG recipient
Michael Chen (2022, Class of 2024), Biological Sciences Major, University of Southern California
Crystal Lovato (2022, Class of 2023), Biological Sciences Major, Idaho State University
2022 SROP student
Andrew Yang (2022, Class of 2025), Math Major, Brown University
2022 Quantitative Biology REU student
Sharik Khan (2022 - 2023, Class of 2024), Biological Sciences Major
2022 Summer URG recipient, 2023 Summer URG recipient
Skyler Stone (2022 - 2024, Class of 2025), Biological Sciences Major
2022 Posner Fellow, 2023 Summer URG recipient
Andrea Phung (2021 - 2024, Class of 2025), Biological Sciences Major
2022 AYURG recipient, 2022 Summer URG recipient
Cassia Yeo (2021 - 2024, Class of 2025), Biological Sciences Major
2022 Summer URG recipient, 2022 Academic Year URG recipient,
2023 Summer Advanced URG recipient, 2023 National Collegiate Undergraduate Research Award
Chloe Sokol (2021 - 2022, Class of 2024), Data Sciences Major
Nicole Banks (2021 - 2022, Class of 2022), Biological Sciences Major
Joey Gallindo (2021, Class of 2024), Engineering Major
2021 Summer URG recipient
Jordan Vaughn (2020 - 2021, Class of 2023), Biological Sciences Major
2021 WCAS Baker Family Summer Grant recipient
Karan Gowda (2020 - 2021, Class of 2022), Biological Sciences Major
2020 Center for Quantitative Biology Summer Grant recipient, 2021 Academic Year URG recipient
Iris Swarthout (2020 - 2021, Class of 2023), Biological Sciences Major
2020 Summer Internship Grant Program recipient, 2021 Summer URG recipient
Raghav Gupta (2020, Class of 2022), Biological Sciences Major
Isabella Miller (2020 - , Class of 2023), Biological Sciences and Hispanic studies double major, Bowdoin Coll.
2022 Summer research grant recipient
Emily Jahn (2019 - 2020, Class of 2020), Biological Sciences Major
2019 Weinberg College Summer Grant recipient
Kailyn Parham (2019 - 2021, Class of 2021), Biological Sciences Major
2019 Summer URG recipient, 2020 Academic Year URG Recipient, 2021 Acad. Year URG Recipient

Anna Derrick (2019 - 2020, Class of 2021), Biological Sciences Major
2019 Weinberg College Summer Grant recipient

Jake Mostad (2018 - 2020, Class of 2020), Biological Sciences Major
2019 Summer URG recipient

Katie Introcaso (2018 - 2022, Class of 2022), Biological Sciences Major
2020 Summer URG recipient, 2021 WCAS Baker Family Summer Grant recipient

Ellen Chao (2018 - 2021, Class of 2021), Biological Sciences Major
2018 Summer URG recipient, 2019 Chemistry of Life Processes Lambert Fellow

Karol Bisaga (2017 - 2019, Class of 2020), Biological Sciences Major
2017 NU Bioscientist Summer Grant recipient

Tim Sheng (2018 - 2019, Class of 2019), Biological Sciences Major
2018 Summer URG recipient

Grace Park (2016 - 2019, Class of 2019), Biological Sciences Major
2016 Posner Fellowship recipient, 2017 Program in Biological Sciences Summer Grant recipient, Natural Sciences and Engineering 2nd place poster prize recipient

Zyneb Adewusi (2018, Class of 2018), Biological Sciences Major, Moraine Valley Community College
2018 Summer Research Opportunities Program (SROP) recipient 2019 Northwestern Undergraduate

Yihong Hu (2018 - 2019, Class of 2021), Biological Sciences Major
2018 Program in Biological Sciences Summer Grant recipient

Chido Chikaturudzi (2017-2018, Class of 2018), Biological Sciences Major, Northeastern Illinois Univ.

Peter Finnegan (2017 - 2018, Class of 2020), Biological Sciences Major
2017 Program in Biological Sciences Summer Grant recipient

Kimberly Collins (2016 - 2018, Class of 2020), Biological Sciences Major
2017 NU Bioscientist Summer Grant recipient

Selina Deiparine (2016 - 2018, Class of 2018), Biological Sciences Major
2016 Summer URG recipient, 2016 Academic URG recipient

Rohit Rastogi (2016 - 2017, Class of 2019), Computer Science and Statistics Majors

Sarah Bier (2016 - 2017, Class of 2019), Biological Sciences Major
2016 Summer URG recipient

Mattlyn Cordova (2016 - 2017, Class of 2019), Gender Studies Major
2016 Program in Biological Sciences Summer Grant recipient

Joshua Roberts (2015-2016, Class of 2016), Computer Science Major

Nicholas Irons (2015, Class of 2018), Physics Major
2015 Summer URG recipient

Annika Zhang (2014-2015, Class of 2018), Biological Sciences Major
2015 Weinberg College Summer Grant recipient

Tyler Shimko (summers 2012, 2013, 2014, 2015, University of Utah Class of 2015), Biology Major
Barry Goldwater Scholarship recipient
Myriad Academic Scholarship recipient
Thomas Verender Hanks Scholarship recipient
National Science Foundation Graduate Research Fellowship recipient
Department of Energy Computational Science Graduate Fellowship Honorable Mention

Mazeed Aro-Lambo (2014, Class of 2017), Biological Sciences Major
2014 NU Bioscientist Summer Grant recipient

Stevie Hippleheuser (2014 - , Class of 2017), Biological Sciences Major
2016 Program in Biological Sciences Summer Grant recipient
2015 Summer URG recipient
2014 Weinberg College Summer Grant recipient

Camille Calvin (2014, Class of 2017), Mechanical Engineering Major
2014 Posner fellowship recipient

Hillary Tsang (2013 - 2016, Class of 2016), Biological Sciences Major

2015 Weinberg Summer Grant recipient

2014 Summer URG recipient

2014 Academic URG recipient

Lautaro Clienti (2013 - 2015, Class of 2017), Mechanical Engineering Major

2014 Academic URG recipient

Kreena Patel (2013 - 2015, Class of 2015), Biological Sciences and Psychology Double Major

2015 Emmanuel Margoliash Prize for Basic Research recipient

Winfred Hill Award recipient

James Alton James Scholar

Ellen Taus Scholarship recipient

J.G. Nolan Scholarship recipient

2014 Academic URG recipient

Zifan Xiang (2014 - 2015, Class of 2015), Biomedical Engineering Major

Stephen Chan (2013 - 2014, Class of 2014), Computer Science Major

2013 Summer URG recipient

Graduate student and post-doctoral advising:

Masters student advising:

Ayeh Khorshidian (Johns Hopkins Genetics and Genomics Masters), 2025

Caroline Bond (Quantitative and Systems Biology Masters), 2021-2022

Anita Huang (Biotechnology Program), 2018 - 2020

Ryan (Heechul) Chung (Quantitative and Systems Biology Masters), 2018 - 2019

Suma Aldakeel (advisor, Cindy Voisine - Northeastern Illinois Univ.), 2016 - 2017

Kristen Larrichia (advisor, Nyree Zerega – Program in Plant Biology and Conservation), 2014 - 2015

Lucie Bastin-Heline (Master's exchange student, Ecole Normale Superior, Paris, France), 2014

Graduate PhD candidates (*Current position*):

Lina Blanco (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Joy Mompi (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Rose Runyan (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Etta Schaye (2025 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Recipient of an Exploration and Field Research Grant for New Zealand field work from The Explorers Club Washington

Maya Mastranardo (2024 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Lance O'Connor (2024 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

NSF GRFP Honorable mention

Recipient of Best TA award - Biochemistry lab (2025)

Ryan McKeown (2022 -), Ph.D. student, Interdisciplinary Biological Sciences Program

Funded by the NIH Biotechnology Training grant full member (2022-2024)

Nic Moya (2020 - 2025), Ph.D. student, Interdisciplinary Biological Sciences Program

Funded by the NIH Biotechnology Training grant full member (2021-2022)

Joy Nyaanga (2019 - 2022), Ph.D. student jointly advised with Niall Mangan, Interdisciplinary Bio. Sci. Program

Funded by NSF-Simons Center for Quantitative Biology (2019-2022)

(Senior scientist, Evozyne)

Loraina Stinson (2019 - 2021), Ph.D. student - left with Masters, Interdisciplinary Biological Sciences Program

Funded by the Cell and Molecular Basis of Disease NIH Training grant (2019-2021)

Clayton Dilks (2018 - 2021), Ph.D. student, Interdisciplinary Biological Sciences Program

Funded by the NIH Biotechnology Training grant cluster (2019)

Funded by the NIH Biotechnology Training grant full member (2020)

Recipient of travel award from WAAVP (2019)

Recipient of Burroughs-Wellcome Travel Award (2019)

(Senior scientist, Evozyne)

Ye Wang (2017 - 2019), Visiting Ph.D. student, Sichuan Agricultural University, China

Funded by China Scholarship Council (2017-2019)

(Program Director for Panda Genomes, the Chengdu Panda Base)

Kathryn Evans (2016 - 2020), Ph.D. student, Interdisciplinary Biological Sciences Program

Funded by the NSF-Simons Center for Quantitative Biology (2020)

Funded by the Cell and Molecular Basis of Disease NIH Training grant (2017-2019)

Recipient of travel awards from IBiS and the Northwestern Graduate School (2017)

Recipient of Biotechnology NIH Training grant cluster member, declined (2017)

(Bioinformaticist, Precision Biosciences, Inc.)

Shannon Brady (2015 - 2019), Ph.D. student, Interdisciplinary Biological Sciences Program

Recipient of the Dr. John N. Nicholson Fellowship (2018-2019)

Funded by the Biotechnology NIH Training grant (2015-2017)

National Science Foundation Graduate Research Fellowship Program (Honorable Mention)

Recipient of travel award from Union Biometrika (2016), IBiS and the NU Graduate School (2017)

Poster first prize winner Northwestern Computational Research Day (2017)

Poster prize winner 21st International C. elegans meeting

Best TA award IBiS Graduate Program (2017)

(Consultant, Boston Consulting Group)

Daniel Cook (2014 - 2018), Ph.D. student, Driskill Graduate Program

Current position: Senior Programmer at Google, Mountain View, CA

Northwestern Graduate School Outstanding Thesis Award (2018)

Funded by a National Science Foundation Pre-doctoral Fellowship (2015-2018)

Northwestern Presidential Fellowship Finalist (2017)

Recipient of travel awards from IBiS and the Northwestern Graduate School (2016)

(Genomics team lead, Google)

Stefan Zdraljevic (2014 - 2019), Ph.D. student, Interdisciplinary Biological Sciences Program

Recipient of the Widom Award for Research Excellence (2019)

Northwestern Graduate School Outstanding Thesis Award (2019)

Recipient of the IBiS Rappaport Award for Research Excellence (2018)

Funded by the Cell and Molecular Basis of Disease NIH Training grant (2015-2017)

Recipient of travel awards from Northwestern Center for Genetic Medicine (2016, 2017)

Recipient of travel awards from IBiS and the Northwestern Graduate School (2015, 2017)

Chemistry of Life Processes Drug Discovery Scholar (2017)

(NIH NRSA F32 post-doctoral fellow at UCLA and HHMI, Kruglyak lab)

Additional rotation graduate students:

Rose Runyan (2026), Lina Blanco (2026), Zhiyu Zheng (2026), Jailyn Loor (2025), Jessica Shaffer (2025), Joy Mompoti (2025), Etta Schaye (2025), Joshua Bauman (2025), Sara Seegers (2025), Izabella Mastroianni (2024), Mingling (Derek) Hui (2024), Tabor Rodriguez (2024), Francesca Batelli (2024), Jake Gavin (2024), Maya Mastronardo (2024), Lance O'Connor (2023), Corinne Croslyn (2022), Ryan McKeown (2022), Brooke Angel (2021), Jack Sumner (2021), Nic Moya (2020), Joy Nyaanga (2019), Loraina Stinson (2019), Emily Czajkowski (2018), Julie Liang (2018), Elan Ness-Cohn (2018), Kyle Siegel (2018), Clayton Dilks (2018), Garth Fisher (2017), Nicholas Sepulveda (2017), Michael Schamber (2017), Evan Buechel (2016), Nic Daffern (2016), Bryan Eder (2016), Ryan Abdella (2015), Erin Baker (2014), Alex Karge (2014), Saiorse McSharry (2014), Amy Nilles (2013), Ian Wolff (2013)

Post-doctoral researchers (Current position):

Connor Leong (2026 -), Ph.D. from Oregon State Univ., advisors Drs. Thomas Sharpton and Michael Kent
 Zihao (John) Li (2026 -), Ph.D. from the University of Pennsylvania, advisor Dr. Chris Fang-Yen
 Nikita Jhaveri (2024 -), Ph.D. from McMaster University, advisor Dr. Bhagwati Gupta

Moore Foundation Post-doctoral grant

Bowen Wang (2024 - 2025), Ph.D. from Northeast Forestry University, advisor Dr. Ling Ma

Amanda Shaver (2022 -), Ph.D. from Univ. of Georgia, advisor Dr. Arthur Edison

Janelia Leading Edge Fellow, 2024

Chair of the National Post-doctoral Council

NIH F32 Post-doctoral fellow (2024-2027)

José Luis Tellez Arreola (2022), Ph.D. from Universidad Autonoma de Mexico, advisor Dr. Ataulfo Torres

(Research Assistant Professor, National University Autonomous of México)

JB Collins (2021 - 2026), Ph.D. from Univ. of Georgia, advisor Dr. Ray Kaplan

USDA Post-doctoral fellow (2024-2027)

(Assistant Professor of Veterinary Clinical and Life Sciences, Utah State University)

Janneke Wit (2019 - 2021), Ph.D. from Aarhus Univ., advisor Dr. Volker Loeschcke

(Research Fellow, Aalborg University)

Sam Widmayer (2019 - 2022), Ph.D. from North Carolina State Univ., advisor Dr. David Aylor

(Associate Computational Scientist, The Jackson Laboratory)

Gaotian Zhang (2018 - 2023), Ph.D. from Ecole Normale Supérieure, Paris, advisor Dr. Marie-Anne Felix

(La Fondation pour la Recherche Médicale, CNRS, Ecole Normale Supérieure - Paris)

Timothy Crombie (2017 - 2023), Ph.D. from University of Florida, advisor Dr. David Julian

(Assistant Professor of Biology, Florida Institute of Technology)

Lewis Stevens (2019 - 2020), Ph.D. from Univ. of Edinburgh, advisor Dr. Mark Blaxter

(Bioinformaticist, Wellcome Trust Sanger Institute, Hinxton, England)

Steffen Hahnel (2017 - 2018), Ph.D. from Justus-Liebig University, advisor Dr. Christoph Grevelding

Recipient of DFG German Science Fellowship (2018-2020)

Recipient of Northwestern Post-doctoral Travel Award (2018)

Recipient of Burroughs-Wellcome Travel Award (2018)

(Scientist, Boehringer Ingelheim Health, Germany)

Daehan Lee (2017 - 2020), Ph.D. from Seoul National University, advisor Dr. Junho Lee

(Assistant Professor of Biological Sciences, Sungkyunkwan University, South Korea)

Mostafa Zamanian (2014 - 2016), Ph.D. from Iowa State University, advisor Dr. Timothy Day

Recipient of NIH/NIAID K22 Career Transition Award (2016)

Recipient of Northwestern Post-doctoral Travel Award (2016)

Funded by the Bill and Melinda Gates Foundation (2014)

(Associate Professor of Pathobiological Sciences, University of Wisconsin - Madison)

Bryn Gaertner (2014), Ph.D. from University of Oregon, advisor Dr. Patrick Phillips

(Associate Scientific Director at Ashfield, part of UDG Healthcare PLC)

Graduate Board Oral (GBO) Exam committee memberships:

2025 (1 Biophysics): Smriti Chibber

2025 (5 CMDDB): Tabor Rodriguez, Olivia Choi, Caroline Moore, Derek Hui, Izabella Mastroianni

2025 (1 CS): Kavya Vaddadi (Chair)

2024 (5 CMDDB): Cameron Allen, Jake Galvin, Betty Huang, Alexis Mottram, Qi Wang

2024 (1 BME): Mao-Jan Lin (Chair)

Graduate thesis committee memberships:

Amanda Ray (advisor, Andrew Gordus) 2025
Calvin Runnels (advisor, Andrew Gordus) 2024 -
Harjit Khaira (advisor, Nichole Broderick) (chair) 2024 -
Mao-Jan Lin (advisor, Ben Langmead) (chair) 2024 -
Nick Wong (advisor, Will Ludington) 2024 -
Sara Carioscia (advisor, Rajiv McCoy) (chair) 2024 - 2025
Corinne Croslyn (advisor, Shelby Blythe) 2022 - 2023 (no longer IBiS preceptor)
Feihong Xu (advisor, Luis Amaral) (chair) 2021 - 2023 (no longer IBiS preceptor)
Elias Guan (advisor, Chris Petersen) (chair) 2021 - 2023 (no longer IBiS preceptor)
Reese Richardson (advisor, Luis Amaral) (chair) 2021 - 2023 (no longer IBiS preceptor)
Idalis Ramirez (advisor, Rick Morimoto) 2021
Taojunfeng Su (advisor, Neil Kelleher) (chair) 2021 - 2023 (no longer IBiS preceptor)
Emily Czajkowski (advisor, Sadie Wignall) 2020 - 2023 (no longer IBiS preceptor)
Elan Ness-Cohn (advisor, Rosemary Braun - DGP, Feinberg School of Medicine) (chair) 2019 - 2021
Gabriel Cavin (advisor, Sadie Wignall) 2019 - 2022
Hannah Horton (advisor, Sadie Wignall) 2019 - 2023
Alex McFarland (advisor, Erica Hartman) (chair) 2018 - 2021
Matt Robey (advisor, Neil Kelleher) 2016 - 2020
Rachel Bakker (advisor, Rich Carthew) 2015 - 2020
Joseph Muldoon (advisors, Neda Bagheri and Josh Leonard) (chair) 2015 - 2020
Aaron Sue (advisor, Thomas O'Halloran) 2014 - 2022
Ritika Giri (advisor, Richard Carthew) 2013 - 2020
Sumach Aldakeel (advisor, Cindy Voisine - Northeastern Illinois Univ.) 2016 - 2017
Adam Hockenberry (advisors, Luis Amaral and Michael Jewett) 2015 - 2017
Sarah Stainbrook (advisor, Keith Tyo) 2015 - 2019
Timothy Toby (advisor, Neil Kelleher) 2015 - 2018
Rose Njoroge (advisor, Sarki Abdulkadir – DGP, Feinberg School of Medicine), 2014 - 2018
Keila Torre-Santiago (advisor, Sadie Wignall) 2014 - 2017
Arianne Rodriguez (advisor, Yun Wang) 2014 (Transferred to DGP)
Lilien Voong (advisor, Alec Wang) 2013 - 2017

DEPARTMENT, COLLEGE, AND UNIVERSITY SERVICE

Johns Hopkins University Department of Biology Service:

2026	Department Vice Chair Faculty chalk talks, organizer Holiday Party organizing committee Dr. Kavi Rangan Faculty Mentorship Committee
2025	Department Vice Chair Faculty chalk talks, organizer Holiday Party organizing committee Dr. Kavi Rangan Faculty Mentorship Committee
2024	Holiday Party organizing committee
2024	Dr. Kavi Rangan Faculty Mentorship Committee Faculty chalk talks, organizer
2023	Faculty search committee, chair Faculty chalk talks, organizer

Johns Hopkins University Krieger School of Arts and Sciences Service:

2026 CMDB Graduate Retreat Organizer
2025 CMDB Executive Committee
KSAS Dissertation Prize Committee
CMDB Graduate Retreat Organizer
CMDB Branding and Web Design Committee
2024 KSAS Dissertation Prize Committee
CMDB Graduate Program Admissions Committee
CMDB Graduate Retreat Organizer
2023 CMDB Graduate Program Admissions Committee

Johns Hopkins University Service:

2026 Review Committee for Innovation Fund for Community-Academic Partnerships
Steering committee for JHURA and ORA merger
Limited Submissions Evaluation Committee
Undergraduate Research Grant (PURA) reviewer
2025 Steering committee for JHURA and ORA merger
Woodrow Wilson Fellowship Review Committee
Limited Submissions Evaluation Committee
Undergraduate Research Grant (PURA) reviewer
2024 Woodrow Wilson Fellowship Review Committee
Ad hoc Pew Scholars Review Committee

Northwestern University Department of Molecular Biosciences Service:

2022 Faculty search committee, chair
2021 Faculty search committee
2019 Faculty search committee
2017 Strategic Planning committee
2016 Faculty search committee
Strategic Planning committee
2015 Faculty search committee
2014 Program Review committee

Northwestern University Weinberg College of Arts and Sciences Service:

2022 WCAS Single-cell Genomics committee
WCAS Teaching Award committee
2019 Program in Biological Sciences curriculum committee
2016 Faculty search committee for the Neurobiology Department
2014 NUIN Post-doc Association, *Interviews and Start-up packages*

Northwestern University Service:

2022 Undergraduate Research Grant review committee
Steering committee for NUSeq Facility in Northwestern Medicine
Ad hoc Limited submissions grant review panel
2021 Undergraduate Research Grant review committee
NSF-Simons Center for Quantitative Biology Pilot grant review panel
Limited Submissions Grant review committee
IBiS Graduate Admissions committee, chair
Steering committee for NUSeq Facility in Northwestern Medicine
IBiS Graduate Student Advisor
Ad hoc reviewer for Data Science Initiative grants and fellowships

2020	Limited submissions grant review panel IBiS Graduate Admissions committee, chair Steering committee for NUSeq Facility in Northwestern Medicine IBiS Graduate Student Advisor
2019	Limited submissions grant review panel NSF-Simons Center for Quantitative Biology Pilot grant review panel IBiS Graduate Admissions committee, chair Steering committee for NUSeq Facility in Northwestern Medicine Masters Program in Quantitative and Systems Biology, Curriculum committee IBiS curriculum committee
2018	IBiS Graduate Admissions committee Steering committee for NUSeq Facility in Northwestern Medicine Masters Program in Quantitative and Systems Biology, Curriculum committee Poster judge, Northwestern Undergraduate Research Symposium <i>Ad hoc</i> reviewer for Chemistry of Life Processes undergraduate grants and fellowships <i>Ad hoc</i> reviewer for Data Science Initiative grants and fellowships NSF-Simons Center for Quantitative Biology Pilot grant review panel
2017	Discussion organizer, Northwestern Computational Research Day Steering committee for NUSeq Facility in Northwestern Medicine <i>Ad hoc</i> reviewer for Chemistry of Life Processes undergraduate grants and fellowships <i>Ad hoc</i> reviewer for Data Science Initiative grants and fellowships
2016	IBiS Computational Bootcamp for incoming graduate students Lurie Cancer Center American Cancer Society IRG review panel Steering committee for NUSeq Facility in Northwestern Medicine <i>Ad hoc</i> reviewer for Chemistry of Life Processes undergraduate grants and fellowships <i>Ad hoc</i> reviewer for Data Science Initiative grants and fellowships
2015	IBiS Graduate Admissions committee IBiS Retreat committee, Co-chair IBiS Computational Bootcamp for incoming graduate students
2014	Creation of IBiS Computational Bootcamp for incoming graduate students IBiS Retreat committee, Co-chair Poster judge, Northwestern Undergraduate Research Symposium IBiS Graduate Admissions committee
2013	Poster judge, Northwestern Undergraduate Research Symposium

COMMUNITY WORK

2023	Hosted 45 8th grade students from North Shore Country Day School for a day of science
2020	STEM presenter at ConnectCon presented by Wizards.exe
2020 - 2025	Assistant Scoutmaster, Troop 2, Wilmette, IL
2015 - 2019	Hosted 80 5th grade students from Lincolnwood Elementary School for a day of science
2014 - 2018	Lecturer on <i>C. elegans</i> genetics to the Latin School of Chicago advanced biology class
2014	Co-organized (with Jacob Cruger) nematode collections with the Punahou School, Hawaii
2009	Organized nematode collections with Vassalboro Community School, Maine